

NEET 2020 Question Paper with Solutions

PHYSICS

Question No. 1**Topic** – Electricity and magnetism**Concept** – Alternating current**Subject Concept** -AC Circuit**Concept Field** – LCR circuit**Question Level** - Hard**Expected time to solve** - 70 sec

1. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :
- (1) 0.5 (2) 1.0 (3) -1.0 (4) zero

Sol. (2)

When L is removed,

$$\tan \phi = \frac{|X_C|}{R} \Rightarrow \tan \frac{\pi}{3} = \frac{X_C}{R} \quad \dots(i)$$

When C is removed,

$$\tan \phi = \frac{|X_L|}{R} \Rightarrow \tan \frac{\pi}{3} = \frac{X_L}{R} \quad \dots(ii)$$

From (i) and (ii), $X_L = X_C$ Since, $X_L = X_C$, the circuit is in resonance. $Z = R$

$$\text{Power factor} = \cos \phi = \frac{R}{Z} = 1$$

Question No. 2**Topic** - Solids**Concept** - Elasticity**Subject Concept** – Hook's Law**Concept Field** – Young's modulus**Question Level** - Moderate**Expected time to solve** - 30 sec

2. A wire of length L , area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :

(1) $\frac{Mg(L_1 - L)}{AL}$ (2) $\frac{MgL}{AL_1}$ (3) $\frac{MgL}{A(L_1 - L)}$ (4) $\frac{MgL_1}{AL}$

Sol. (3)

$$\text{Stress} = \frac{Mg}{A}$$

$$\text{Strain} = \frac{\Delta L}{L} = \frac{L_1 - L}{L}$$

$$\text{Young's modulus} = \frac{\text{Stress}}{\text{Strain}} = \frac{MgL}{A(L_1 - L)}$$

Question No. 3

Topic – Electricity and magnetism

Concept – Magnetic effect of current

Subject Concept – Ampere's circuital law

Concept Field – Magnetic field due to solenoid

Question Level - easy

Expected time to solve - 40 sec

3. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :

$$(\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1})$$

- (1) $3.14 \times 10^{-4} \text{ T}$ (2) $6.28 \times 10^{-5} \text{ T}$
 (3) $3.14 \times 10^{-5} \text{ T}$ (4) $6.28 \times 10^{-4} \text{ T}$

Sol. (4)

Magnetic field at centre of solenoid = $\mu_0 n l$

$$n = \frac{N}{L} = \frac{100}{50 \times 10^{-2}} = 200 \text{ turns/m}$$

$$l = 2.5 \text{ A}$$

On putting the values

$$B = 4\pi \times 10^{-7} \times 200 \times 2.5$$

$$= 6.28 \times 10^{-4} \text{ T}$$

Question No. 4

Topic - Optics

Concept – Ray optics

Subject Concept - Prism

Concept Field – Refraction through prism

Question Level - easy

Expected time to solve - 40 sec

4. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , then the angle of incidence is nearly equal to :

(1) $\frac{2A}{\mu}$

(2) μA

(3) $\frac{\mu A}{2}$

(4) $\frac{A}{2\mu}$

Sol. (2)

Light ray emerges normally from another surface, hence, e (angle of emergence) = 0

$$r_2 = 0$$

$$r_1 + r_2 = A$$

$$\Rightarrow r_1 = A$$

Applying Snell's law on first surface

$$1 \cdot \sin i = \mu \sin r_1$$

$$\Rightarrow \sin i = \mu \sin A$$

For small angles ($\sin \theta \approx \theta$)

hence $i = \mu A$

Question No. 5

Topic – Electricity and magnetism

Concept - Electrostatics

Subject Concept – Electric potential

Concept Field – Relation between Electric potential and field

Question Level - Easy

Expected time to solve - 30 sec

5. In a certain region of space with volume 0.2 m^3 , the electric potential is found to be 5V throughout. The magnitude of electric field in this region is :

(1) 0.5 N/C

(2) 1 N/C

(3) 5 N/C

(4) zero

Sol. (4)

Since, electric potential is found throughout constant, hence electric field,

$$E = -\frac{dV}{dr} = 0$$

Question No. 6

Topic -Modern physics

Concept – Atom

Unacademy Select

Subject Concept – Bohr's model

Concept Field – Drawbacks of Bohr's model

Question Level – easy

Expected time to solve – 30 sec

6. For which one of the following, Bohr model is not valid ?

- (1) Singly ionised helium atom (He)
- (2) Deuteron atom
- (3) Singly ionised neon atom (Ne)
- (4) Hydrogen atom

Sol. (3)

Bohr model is only valid for single electron species. Singly ionised neon atom has more than one electron in orbit. Hence, Bohr model is not valid.

Question No. 7

Topic – Modern physics

Concept – Photo electric effect

Subject Concept – Property of photon

Concept Field – Intensity

Question Level – Easy

Expected time to solve – 20 sec

7. Light with an average flux of 20 W/cm falls on a non-reflecting surface at normal incidence having surface area 20 cm². The energy received by the surface during time span of 1 minute is :

- (1) 12×10^3 J
- (2) 24×10^3 J
- (3) 48×10^3 J
- (4) 10×10^3 J

Sol. (2)

Energy received = Intensity $\times$ Area $\times$ Time
= $20 \times 20 \times 60$
= 24×10^3 J

Question No. 8

Topic – Modern physics

Concept – Dual nature of light

Subject Concept – De Broglie Equation

Concept Field – 3

Question Level – Easy

Expected time to solve – 30 sec

8. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is

- (1) 10^2 V
- (2) 10^3 V
- (3) 10^4 V
- (4) 10 V

Sol. (3)

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$\sqrt{V} = \frac{12.27 \times 10^{-10}}{1.227 \times 10^{-11}} = 10^2$$

$$\therefore V = 10^4 \text{ volts}$$

Question No. 9

Topic - Mechanics

Concept - Gravitation

Subject Concept – Gravitational force

Concept Field – Gravitational acceleration at h height from earth's surface

Question Level - moderate

Expected time to solve - 40 sec

9. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth?

- (1) 32 N (2) 30 N (3) 24 N (4) 48 N

Sol. (1)

$$mg_h = \frac{mg_0}{\left(1 + \frac{h}{R}\right)^2}$$

$$W = \frac{72}{\left(1 + \frac{R/2}{R}\right)^2}$$

$$W = \frac{72}{(3/2)^2} = \frac{4}{9} \times 72 = 32 \text{ N}$$

Question No. 10

Topic – Electricity and magnetism

Concept - Current Electricity

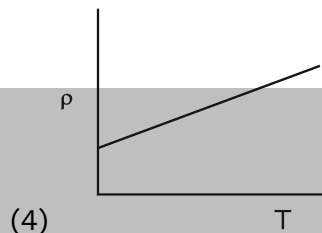
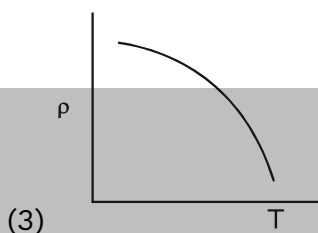
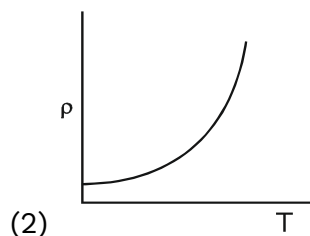
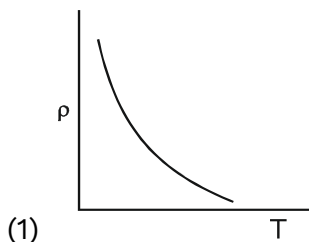
Subject Concept – Conducting material

Concept Field – Curve between resistivity and temperature

Question Level - Easy

Expected time to solve - 20 sec

10. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?



Sol. (2)

At temperature much lower than 0°C , graph deviates considerably from a straight line. Option (2) is correct.

Question No. 11

Topic – Modern physics

Concept – Dual nature of light

Subject Concept – Photo electric effect

Concept Field – Effect of applied frequency on photo electric current

Question Level – Easy

Expected time to solve – 30 sec

11. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled?

- (1) four times (2) one-fourth (3) zero (4) doubled

Sol. (3)

$$\nu = \frac{3}{2} \nu_0$$

$$\nu' = \frac{\nu}{2} = \frac{3}{4} \nu_0$$

$$\therefore \nu' < \nu_0$$

$\therefore$ No photoelectric emission will take place.

Question No. 12

Topic – Electricity and magnetism

Concept – Alternating current

Subject Concept – alternating circuits

Concept Field – pure capacitance circuit

Question Level - Easy

Expected time to solve - 30 sec

12. A $40\ \mu\text{F}$ capacitor is connected to a $200\ \text{V}$, $50\ \text{Hz}$ ac supply. The rms value of the current in the circuit is, nearly:

- (1) $2.05\ \text{A}$ (2) $2.5\ \text{A}$ (3) $25.1\ \text{A}$ (4) $1.7\ \text{A}$

Sol. (2)

$$i_{\text{rms}} = C\omega \varepsilon_{\text{rms}}$$

$$C = 40 \times 10^{-6}\text{F}$$

$$\omega = 2\pi f = 100\pi$$

$$\varepsilon_{\text{rms}} = 200\text{V}$$

$$\begin{aligned}\therefore i_{\text{rms}} &= 200 \times 40 \times 10^{-6} \times 2\pi \times 50 \\ &= 2.5\ \text{A}\end{aligned}$$

Question No. 13

Single Correct Answer

Topic - Optics

Concept – wave optics

Subject Concept – resolving power

Concept Field – resolving power of telescope

Question Level - easy

Expected time to solve - 40 sec

13. Assume that light of wavelength $600\ \text{nm}$ is coming from a star. The limit of resolution of telescope whose objective has a diameter of $2\ \text{m}$ is :

- (1) $1.83 \times 10^{-7}\ \text{rad}$ (2) $7.32 \times 10^{-7}\ \text{rad}$ (3) $6.00 \times 10^{-7}\ \text{rad}$ (4) $3.66 \times 10^{-7}\ \text{rad}$

Sol. (4)

$$\theta_R = 1.22 \frac{\lambda}{d}; \lambda = 600 \times 10^{-9}\text{m}, d = 2\text{m}$$

$$= \frac{1.22 \times 600 \times 10^{-9}}{2}$$

$$\theta = 3.66 \times 10^{-7}\text{rad}$$

Question No. 14

Single Correct Answer

Topic - kinematics

Concept – 1-d motion

Subject Concept – uniform acceleration

Concept Field – motion under gravity

Question Level - easy

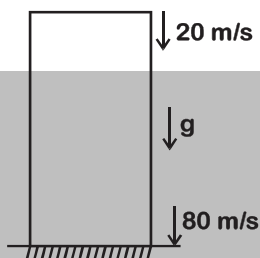
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Expected time to solve - 50 sec

- 14.** A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is : ($g = 10 \text{ m/s}^2$)

(1) 340 m (2) 320 m (3) 300 m (4) 360 m

Sol. (3)



$$v^2 = u^2 + 2gh$$

$$v = 80 \text{ m/s}$$

$$u = 20 \text{ m/s}$$

$$h = \frac{v^2 - u^2}{2g} = \frac{6400 - 400}{20} = 300 \text{ m}$$

Question No. 15

Single Correct Answer

Topic – thermal physics

Concept – kinetic theory of gases

Subject Concept – ideal gas

Concept Field – ideal gas equation

Question Level - moderate

Expected time to solve - 50 sec

- 15.** A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is : ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

(1) 0.2 kg/m^3 (2) 0.1 kg/m^3 (3) 0.02 kg/m^3 (4) 0.5 kg/m^3

Sol. (1)

$$PM = \rho RT \Rightarrow \rho = \frac{PM}{RT}$$

$$P = 249 \times 10^3 \text{ N/m}^2$$

$$M = 2 \times 10^{-3} \text{ kg}$$

$$T = 300 \text{ K}$$

$$\therefore \rho = \frac{(249 \times 10^3)(2 \times 10^{-3})}{8.3 \times 300} = \frac{0.2 \text{ kg}}{\text{m}^3}$$

Question No. 16**Single Correct Answer****Topic – modern physics****Concept – nuclear physics****Subject Concept – nuclear reaction****Concept Field – nuclear fission****Question Level – easy****Expected time to solve – 40 sec**

16. When a uranium isotope ${}_{92}^{235}\text{U}$ is bombarded with a neutron, it generates ${}_{36}^{89}\text{Kr}$, three neutrons and :

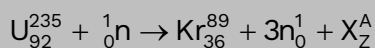
(1) ${}_{40}^{91}\text{Zr}$

(2) ${}_{36}^{101}\text{Kr}$

(3) ${}_{36}^{103}\text{Kr}$

(4) ${}_{56}^{144}\text{Br}$

Sol. (4)



$$92 + 0 = 36 + Z$$

$$\Rightarrow Z = 56$$

$$235 + 1 = 89 + 3 + A$$

$$\Rightarrow A = 144$$

So, ${}_{56}^{144}\text{Ba}$ is generated.

Question No. 17**Single Correct Answer****Topic – modern physics****Concept – semiconductor****Subject Concept – p-n junction****Concept Field – p-n junction biasing****Question Level – easy****Expected time to solve – 20 sec**

17. The increase in the width of the depletion region in a p-n junction diode is due to :

(1) reverse bias only

(2) both forward bias and reverse bias

(3) increase in forward current

(4) forward bias only

Sol. (1)

Due to reverse biasing, the width of the depletion region increases.

Question No. 18**Single Correct Answer****Topic – mechanics****Unacademy Select**

Concept – oscillation

Subject Concept - simple harmonic motion

Concept Field – displacement and acceleration in SHM

Question Level - easy

Expected time to solve - 30 sec

18. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :

- (1) $\frac{3\pi}{2}$ rad (2) $\frac{\pi}{2}$ rad (3) zero (4) π rad

Sol. (4)

$$\text{If } y = A \sin \omega t$$

$$\text{then } v = \frac{dy}{dt}$$

$$v = A\omega \cos \omega t$$

$$a = \frac{dv}{dt}$$

$$a = -A\omega^2 \sin(\omega t)$$

$$a = A\omega^2 \sin(\omega t + \pi)$$

So, phase difference between displacement and acceleration is π .

Question No. 19

Single Correct Answer

Topic – electricity and magnetism

Concept - magnetism

Subject Concept – magnetic material

Concept Field – magnetic susceptibility

Question Level - moderate

Expected time to solve - 50 sec

19. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is :

$$(\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1})$$

(1) $8.0 \times 10^{-5} \text{ T m A}^{-1}$

(2) $2.4 \pi \times 10^{-5} \text{ T m A}^{-1}$

(3) $2.4 \pi \times 10^{-7} \text{ T m A}^{-1}$

(4) $2.4 \pi \times 10^{-4} \text{ T m A}^{-1}$

Sol. (4)

$$\chi_m = 599$$

$$\mu_r = 1 + \chi_m = 600$$

$$\mu = \mu_r \mu_0$$

$$\mu = 600 \times 4\pi \times 10^{-7}$$

$$\mu = 2400\pi \times 10^{-7}$$

$$\mu = 2.4\pi \times 10^{-4} \text{ T m A}^{-1}$$

Question No. 20

Single Correct Answer

Topic – thermal physics

Concept – heat transfer

Subject Concept – heat transfer in spherical object

Concept Field – heat transfer formula for sphere

Question Level – moderate

Expected time to solve – 50 sec

20. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5 r_2$) through 1 K are in the ratio :

- (1) $\frac{9}{4}$ (2) $\frac{3}{2}$ (3) $\frac{5}{3}$ (4) $\frac{27}{8}$

Sol. (4)

$$\Delta Q = ms\Delta T$$

$$\Delta Q = \frac{4}{3} \pi r^3 \rho s \Delta T$$

$$\begin{aligned} \frac{\Delta Q_1}{\Delta Q_2} &= \left(\frac{r_1}{r_2} \right)^3 \\ &= (1.5)^3 \\ &= \frac{27}{8} \end{aligned}$$

Question No. 21

Single Correct Answer

Topic – mechanics

Concept – centre of mass

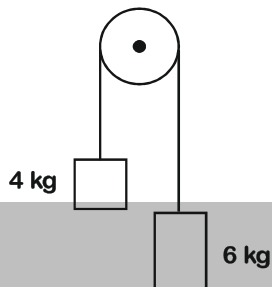
Subject Concept – constrained motion

Concept Field – acceleration block in pulley system

Question Level – Hard

Expected time to solve – 50 sec

21. A Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is :



- (1) $g/2$ (2) $g/5$ (3) $g/10$ (4) g

Sol. (2)

$$a = \frac{(m_1 - m_2)g}{(m_1 + m_2)} \text{ where } m_1 > m_2$$

$$a = \frac{(6 - 4)g}{6 + 4}$$

$$a = \frac{g}{5}$$

Note : Here no option is given according to acceleration of COM of the system.

Question No. 22

Single Correct Answer

Topic – thermal physics

Concept – kinetic theory of gases

Subject Concept – mean free path

Concept Field – relation between molecular diameter and number density

Question Level - easy

Expected time to solve - 20 sec

22. The mean free path for a gas, with molecular diameter d and number density n can be expressed as :

(1) $\frac{1}{\sqrt{2}n\pi d^2}$

(2) $\frac{1}{\sqrt{2}n^2\pi d^2}$

(3) $\frac{1}{\sqrt{2}n^2\pi^2 d^2}$

(4) $\frac{1}{\sqrt{2}n\pi d}$

Sol. (1)

According to the formula

$$\lambda = \frac{1}{\sqrt{2}n\pi d^2}$$

Question No. 23**Single Correct Answer****Topic – electricity and magnetism****Concept** - electrostatics**Subject Concept** -electric dipole**Concept Field** –potential due to electric dipole**Question Level** – easy**Expected time to solve** - 30 sec

- 23.** A short electric dipole has a dipole moment of 16×10^{-9} C m. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :

$$\left(\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \right)$$

- (1) 200 V (2) 400 V (3) zero (4) 50 V

Sol. (1)

$$V = \frac{k p \cos \theta}{r^2}$$

$$V = \frac{9 \times 10^9 \times 16 \times 10^{-9} \times \cos 60}{0.36}$$

$$V = 200 \text{ V}$$

Question No. 24**Single Correct Answer****Topic - solids****Concept** - elasticity**Subject Concept** -stress**Concept Field** – dimension of stress**Question Level** - easy**Expected time to solve** - 20 sec

- 24.** A Dimensions of stress are :

- (1) $[ML^2T^{-2}]$ (2) $[ML^0T^{-2}]$ (3) $[ML^{-1}T^{-2}]$ (4) $[MLT^{-2}]$

Sol. (3)

$$\text{Stress} = \frac{\text{Force}}{\text{Area}}$$

$$= \frac{[MLT^{-2}]}{[L^2]}$$

$$= [ML^{-1}T^{-2}]$$

Question No. 25**Single Correct Answer****Topic - mechanics****Concept** – work, energy, power**Subject Concept** - energy**Concept Field** – relation in joule and electron volt**Question Level** - easy**Expected time to solve** - 30 sec

- 25.** The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly
(1) 0.6 (2) 0.06 (3) 0.006 (4) 6

Sol. (2)

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ J} = \frac{1}{1.6 \times 10^{-19}} \text{ eV}$$

$$\begin{aligned} 10^{-20} \text{ J} &= \frac{10^{-20}}{1.6 \times 10^{-19}} \text{ eV} \\ &= 0.06 \text{ eV} \end{aligned}$$

Question No. 26**Single Correct Answer****Topic – wave motion****Concept** – musical instruments**Subject Concept** - beat**Concept Field** – beat formula**Question Level** - easy**Expected time to solve** - 30 sec

- 26.** In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be
(1) 524 Hz (2) 536 Hz (3) 537 Hz (4) 523 Hz

Sol. (1)

Difference of f_A and f_B is 6 Hz

If tension decreases, f_B decreases and becomes f'_B .

Now, difference of f_A and $f'_B = 7$ Hz (increases)

So, $f_A > f'_B$

$$f_A - f_B = 6 \text{ Hz}$$

$$f_A = 530\text{Hz}$$

$$f_B = 524\text{Hz (original)}$$

Question No. 27

Single Correct Answer

Topic – electricity and magnetism

Concept – current electricity

Subject Concept – types of resistors

Concept Field – Carbon resistor

Question Level – easy

Expected time to solve – 30 sec

27. The color code of a resistance is given below



The values of resistance and tolerance, respectively, are

- (1) 47 k Ω , 10% (2) 4.7 k Ω , 5% (3) 470 Ω , 5% (4) 470 k Ω , 5%

Sol. (3)

According to colour coding

Yellow = 4; Violet = 7; Brown = 1; Gold = 5%

$$\text{So, } R = 47 \times 10^1 \pm 5\%$$

$$R = 470 \pm 5\% \Omega$$

Question No. 28

Single Correct Answer

Topic – optics

Concept – wave optics

Subject Concept – polarization of light

Concept Field – Brewster's law

Question Level – easy

Expected time to solve – 3 sec

28. The Brewster's angle i_b for an interface should be

- (1) $30^\circ < i_b < 45^\circ$ (2) $45^\circ < i_b < 90^\circ$ (3) $i = 90^\circ$ (4) $0^\circ < i_b < 30^\circ$

Sol. (2)

$$\mu = \tan i_b$$

$$1 < \mu < \infty$$

$$1 < \tan i_b < \infty$$

$$\tan^{-1}(1) < i_b < \tan^{-1}(\infty)$$

$$45^\circ < i_b < 90^\circ$$

Question No. 29

Single Correct Answer

Topic – electricity and magnetism

Concept - capacitance

Subject Concept – parallel plate capacitor

Concept Field – effect of medium on capacitance

Question Level - easy

Expected time to solve - 30 sec

29. The capacitance of a parallel plate capacitor with air as medium is $6 \mu\text{F}$. With the introduction of a dielectric medium, the capacitance becomes $30 \mu\text{F}$. The permittivity of the medium is :

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2})$$

$$(1) 1.77 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$(2) 0.44 \times 10^{-10} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$(3) 5.00 \text{C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$(4) 0.44 \times 10^{-13} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$$

Sol. (2)

$$C = KC_0$$

$$K = \frac{C}{C_0} = \frac{30}{6} = 5$$

$$K = \frac{\epsilon}{\epsilon_0}$$

$$\epsilon = K\epsilon_0$$

$$= 5 \times 8.85 \times 10^{-12}$$

$$= 0.44 \times 10^{-10} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$$

Question No. 30

Single Correct Answer

Topic - mechanics

Concept – rigid dynamics

Subject Concept - rotational motion

Concept Field - torque

Question Level - easy

Expected time to solve - 40 sec

- 30.** Find the torque about the origin when a force of $3\hat{j}\text{N}$ acts on a particle whose position vector is $2\hat{k}\text{m}$.

(1) $6\hat{j}\text{ Nm}$ (2) $-6\hat{i}\text{ Nm}$ (3) $6\hat{k}\text{ Nm}$ (4) $6\hat{i}\text{ Nm}$

Sol. (2)

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = 2\vec{k} \times 3\vec{j}$$

$$\vec{\tau} = -6\vec{i}\text{ Nm}$$

Question No. 31

Single Correct Answer

Topic – electricity and magnetism

Concept - Current Electricity

Subject Concept – electric circuits

Concept Field – meter bridge

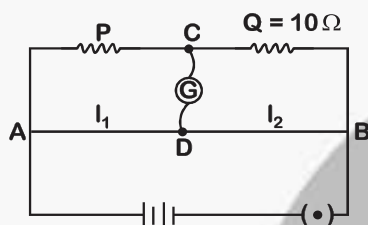
Question Level - hard

Expected time to solve - 70 sec

- 31.** A resistance wire connected in the left gap of a metre bridge balances a $10\ \Omega$ resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of $1\ \Omega$ of the resistance wire is :

(1) $1.0 \times 10^{-1}\text{ m}$ (2) $1.5 \times 10^{-1}\text{ m}$ (3) $1.5 \times 10^{-2}\text{ m}$ (4) $1.0 \times 10^{-2}\text{ m}$

Sol. (1)



$$\text{Initially, } \frac{P}{10} = \frac{l_1}{l_2} = \frac{3}{2}$$

$$\Rightarrow P = \frac{30}{2} = 15\Omega$$

$$\text{Now resistance, } R = \frac{\rho l}{A}$$

$$\frac{R_1}{R_2} = \frac{l_1}{l_2}$$

$$\Rightarrow \frac{15}{1} = \frac{1.5}{l_2}$$

$$l_2 = 0.1 \text{ m}$$

$$= 1.0 \times 10^{-1} \text{ m}$$

Question No. 32

Single Correct Answer

Topic – modern physics

Concept - semiconductor

Subject Concept – transistor

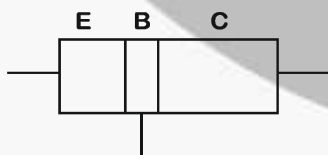
Concept Field – formation of transistor

Question Level - easy

Expected time to solve - 30 sec

- 32.** For transistor action, which of the following statements is correct ?
- (1) Base, emitter and collector regions should have same size.
 - (2) Both emitter junction as well as the collector junction are forward biased.
 - (3) The base region must be very thin and lightly doped.
 - (4) Base, emitter and collector regions should have same doping concentrations.

Sol. (3)



For Bi-polar junction transistor

Length Profile is $L_C > L_E > L_B$

and doping profile is $E > C > B$

For transistor action Base-emitter junction is forward biased and Base-collector junction is reversed biased.

Question No. 33

Single Correct Answer

Topic – electricity and magnetism

Concept – electromagnetic waves

Subject Concept – properties of electromagnetic waves

Concept Field – ratio of electric and magnetic field in electromagnetic waves

Question Level - easy

Expected time to solve - 30 sec

- 33.** The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is :
- (c = speed of electromagnetic waves)

(1) 1 : 1

(2) 1 : c

(3) 1 : c²

(4) c : 1

Sol. (1)

In an electromagnetic wave, half of the intensity is provided by the electric field and half by the magnetic field.

Hence required ratio should be 1 : 1.

Question No. 34

Single Correct Answer

Topic – electricity and magnetism

Concept - Current Electricity

Subject Concept – current flow in conductors

Concept Field - mobility

Question Level - easy

Expected time to solve - 20 sec

34. A charged particle having drift velocity of $7.5 \times 10^{-4} \text{ ms}^{-1}$ in an electric field of $3 \times 10^{-10} \text{ Vm}^{-1}$, has a mobility in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$ of :

(1) 2.5×10^8

(2) 2.5×10^{-6}

(3) 2.25×10^{-15}

(4) 2.25×10^{15}

Sol. (1)

$$\text{Mobility, } \mu = \frac{V_d}{E}$$

$$= \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}}$$

$$= 2.5 \times 10^6 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

Question No. 35

Single Correct Answer

Topic – electricity and magnetism

Concept - electrostatics

Subject Concept – electric field

Concept Field - electric field due to charged sphere

Question Level - easy

Expected time to solve - 40 sec

35. A spherical conductor of radius 10 cm has a charge of $3.2 \times 10^{-7} \text{ C}$ distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere?

$$\left(\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \right)$$

(1) $1.28 \times 10^5 \text{ N/C}$

(2) $1.28 \times 10^6 \text{ N/C}$

(3) $1.28 \times 10^7 \text{ N/C}$

(4) $1.28 \times 10^4 \text{ N/C}$

Sol. (1)

Electric field outside a conducting sphere

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \\ &= \frac{9 \times 10^9 \times 3.2 \times 10^{-7}}{225 \times 10^{-4}} \\ &= 0.128 \times 10^6 \\ &= 1.28 \times 10^5 \text{ N/C} \end{aligned}$$

Question No. 36

Single Correct Answer

Topic – basics mathematics in physics

Concept – significant number

Subject Concept – properties

Concept Field – significant figures rules

Question Level – easy

Expected time to solve – 30 sec

36. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$?

- (1) 9.98 m (2) 9.980 m (3) 9.9 m (4) 9.9801 m

Sol. (1)

$$\begin{array}{r} 9.99 \\ - 0.0099 \\ \hline 9.9801 \text{ m} \end{array}$$

In subtraction, answer should be reported by least number of decimal places, so answer should be 9.98 m.

Question No. 37

Single Correct Answer

Topic – optics

Concept – wave optics

Subject Concept – interference

Concept Field – young's double slit experiment

Question Level – moderate

Expected time to solve – 40 sec

37. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes:

- (1) half (2) four times (3) one-fourth (4) double

Sol. (2)

$$\text{Fringe width } \beta = \frac{\lambda D}{d}$$

$$\text{Now, } d' = \frac{d}{2} \text{ and } D' = 2D$$

$$\text{So, } \beta' = \frac{\lambda(2D)}{d/2} = \frac{4\lambda D}{d}$$

$$\beta' = 4\beta$$

Question No. 38

Single Correct Answer

Topic – thermal physics

Concept - thermodynamics

Subject Concept – thermodynamic process

Concept Field – adiabatic process

Question Level - easy

Expected time to solve - 20 sec

38. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is:

- (1) adiabatic (2) isochoric (3) isobaric (4) isothermal

Sol. (1)

Entire system is thermally insulated. So, no heat exchange will take place. Hence, process will be adiabatic.

Question No. 39

Single Correct Answer

Topic – modern physics

Concept – nuclear physics

Subject Concept – energy

Concept Field – energy- mass equivalence

Question Level - easy

Expected time to solve - 50 sec

39. The energy equivalent of 0.5 g of a substance is :

- (1) 4.5×10^{13} J (2) 1.5×10^{13} J (3) 0.5×10^{13} J (4) 4.5×10^{16} J

Sol. (1)

From mass-energy equivalence.

$$\begin{aligned}
 E &= mc^2 \\
 &= 0.5 \times 10^{-3} \times (3 \times 10^8)^2 \\
 &= 4.5 \times 10^{13} \text{ J}
 \end{aligned}$$

Question No. 40

Single Correct Answer

Topic – fluid mechanics

Concept – mechanical properties of fluid

Subject Concept – surface tension

Concept Field – capillary effect

Question Level - modrate

Expected time to solve - 50 sec

40. A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of the water in the capillary is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is :

- (1) 5.0 g (2) 10.0 g (3) 20.0 g (4) 2.5 g

Sol. (2)

Force of surface tension balances the weight of water in capillary tube.

$$F_s = 2\pi r T \cos \theta = mg$$

Here, T and θ are constant

So, $m \propto r$

$$\text{Hence, } \frac{m_2}{5.0} = \frac{2r}{r}$$

$$\Rightarrow m_2 = 10.0 \text{ g}$$

Question No. 41

Single Correct Answer

Topic – basics and instruments

Concept – measuring instruments

Subject Concept – screw gauge

Concept Field – least count

Question Level - easy

Expected time to solve - 40 sec

41. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :

- (1) 0.25 mm (2) 0.5 mm (3) 1.0 mm (4) 0.01 mm

Sol. (2)

$$\text{Least count} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

$$0.01 \text{ mm} = \frac{\text{Pitch}}{50}$$

$$\text{Pitch} = 0.5 \text{ mm}$$

Question No. 42

Single Correct Answer

Topic – modern physics

Concept - semiconductor

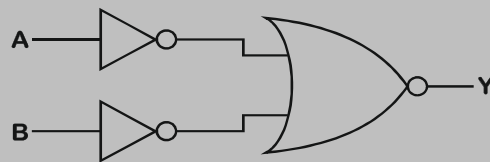
Subject Concept – logic gates

Concept Field – not and nor gate

Question Level - moderate

Expected time to solve - 50 sec

42. For the logic circuit shown, the truth table is:



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(2)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

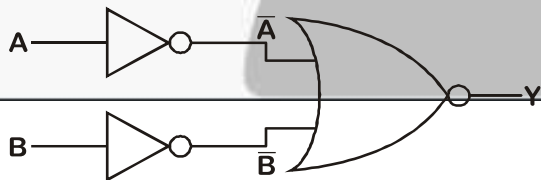
(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Sol. (4)



$$Y = \overline{\overline{A} + \overline{B}}$$

$$= \overline{\overline{A \cdot B}}$$

$$= A \cdot B \Rightarrow \text{AND Gate}$$

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Question No. 43

Single Correct Answer

Topic – thermal physics

Concept – kinetic theory of gases

Subject Concept – degree of freedom

Concept Field – average kinetic energy

Question Level – easy

Expected time to solve – 40 sec

43. The average thermal energy for a mono-atomic gas is: (k_B is Boltzmann constant and T , absolute temperature)

- (1) $\frac{3}{2}k_B T$ (2) $\frac{5}{2}k_B T$ (3) $\frac{7}{2}k_B T$ (4) $\frac{1}{2}k_B T$

Sol. (1)

For monoatomic gases, degree of freedom is 3. Hence average thermal energy per molecule is $KE_{avg} = \frac{3}{2}dk_B T$.

Question No. 44

Single Correct Answer

Topic – electricity and magnetism

Concept – current electricity

Subject Concept – effect of temperature on materials

Concept Field – temperature coefficient of resistance

Question Level – easy

Expected time to solve – 20 sec

44. The solids which have the negative temperature coefficient of resistance are:

- (1) insulators only (2) semiconductors only
(3) insulators and semiconductors (4) metals

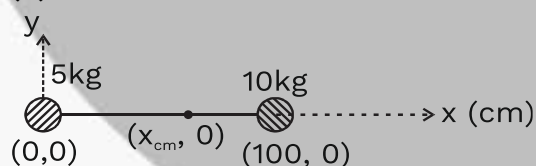
Sol. (3)

For metals temperature coefficient of resistance is positive while for insulators and semiconductors, temperature coefficient of resistance is negative.

Question No. 45**Single Correct Answer****Topic – mechanics****Concept – rigid body****Subject Concept – centre of mass****Concept Field – position of centre of mass****Question Level – moderate****Expected time to solve – 60 sec**

45. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of:

(1) 50 cm (2) 67 cm (3) 80 cm (4) 33 cm

Sol. (2)

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$
$$= \frac{5 \times 0 + 10 \times 100}{5 + 10} = \frac{200}{3} = 66.66 \text{ cm}$$
$$x_{cm} = 67 \text{ cm}$$

NEET UG 2020 Oct 14 Chemistry

Question Type: NEET**Difficulty of question: Easy****Expected time to solve: 20 sec.****Topic: Environmental Chemistry****Concept: Environmental Chemistry****Sub concept: Acid rain****Concept field: Acid rain**

1. Which of the following statement is NOT true about acid rain?

- (1) It is due to reaction of SO_2 , NO_2 and CO_2 with rain water.
- (2) Causes no damage to monuments like Taj Mahal.
- (3) It is harmful for plants.
- (4) Its pH is less than 5.6

Sol. (2)

Acid rain causes to damage to monuments like Taj mahal.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Basic of Organic Chemistry

Concept: Oxidation Reduction

Sub concept: Oxidation Number

Concept field: Oxidation Number

2. The oxidation number of the underlined atom in the following species.

- (1) $\text{Cu}_2\underline{\text{O}}$ is -1
- (2) $\underline{\text{Cl}}\text{O}_3^-$ is +5
- (3) $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$ is +6
- (4) $\text{H}\underline{\text{Au}}\text{Cl}_4$ is +3

Identify the incorrect option

Sol. (1)

Oxidation state of "O" in Cu_2O = -2

Oxidation state of "Cl" in ClO_3^- = +5

Oxidation state of "Cr" in $\text{K}_2\text{Cr}_2\text{O}_7$ = +6

Oxidation state of "Au" in HAuCl_4 = +3

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Amines

Concept: Hoffmann bromamide degradation reaction

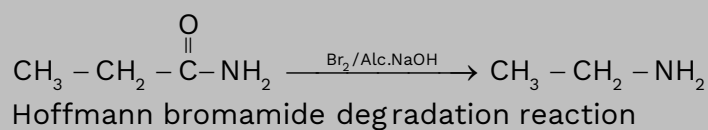
Sub concept: Hoffmann bromamide degradation reaction

Concept field: In presence of X_2 /alcoholic NaOH

3. Reaction of propanamide with ethanolic sodium hydroxide and bromine will give

- | | |
|-----------------|-----------------|
| (1) Ethylamine | (2) Methylamine |
| (3) Propylamine | (4) Aniline |

Sol. (1)



Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 sec.

Topic: Mole Concepts

Concept: Purification or Separation

Sub concept: Separation Analysis

Concept field: Separation Analysis

4. A liquid compound (x) can be purified by steam distillation only if it is

- (1) Steam volatile, immiscible with water
- (2) Not steam volatile, miscible with water
- (3) Steam volatile, miscible with water
- (4) Not steam volatile, immiscible with water

Sol. (1)

Compounds purified by steam distillation which are immiscible in water but steam volatile.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 sec.

Topic: Chemical Bonding

Concept: Hybridization

Sub concept: Geometry

Concept field: Geometry

5. Among the compounds shown below which one revealed a linear structure

(1) NO_2

(2) HOCl

(3) O_3

(4) N_2O

Sol. (4) $:\text{N} \equiv \text{N} \rightarrow \ddot{\text{O}}: (\text{Linear})$

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 Sec

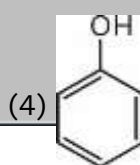
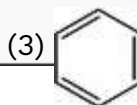
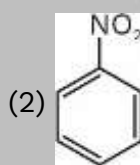
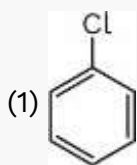
Topic: Hydrocarbon

Concept: Aromatic Hydrocarbon

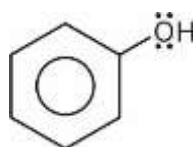
Sub concept: Electrophilic substitution reaction

Concept field: Electrophilic substitution reaction

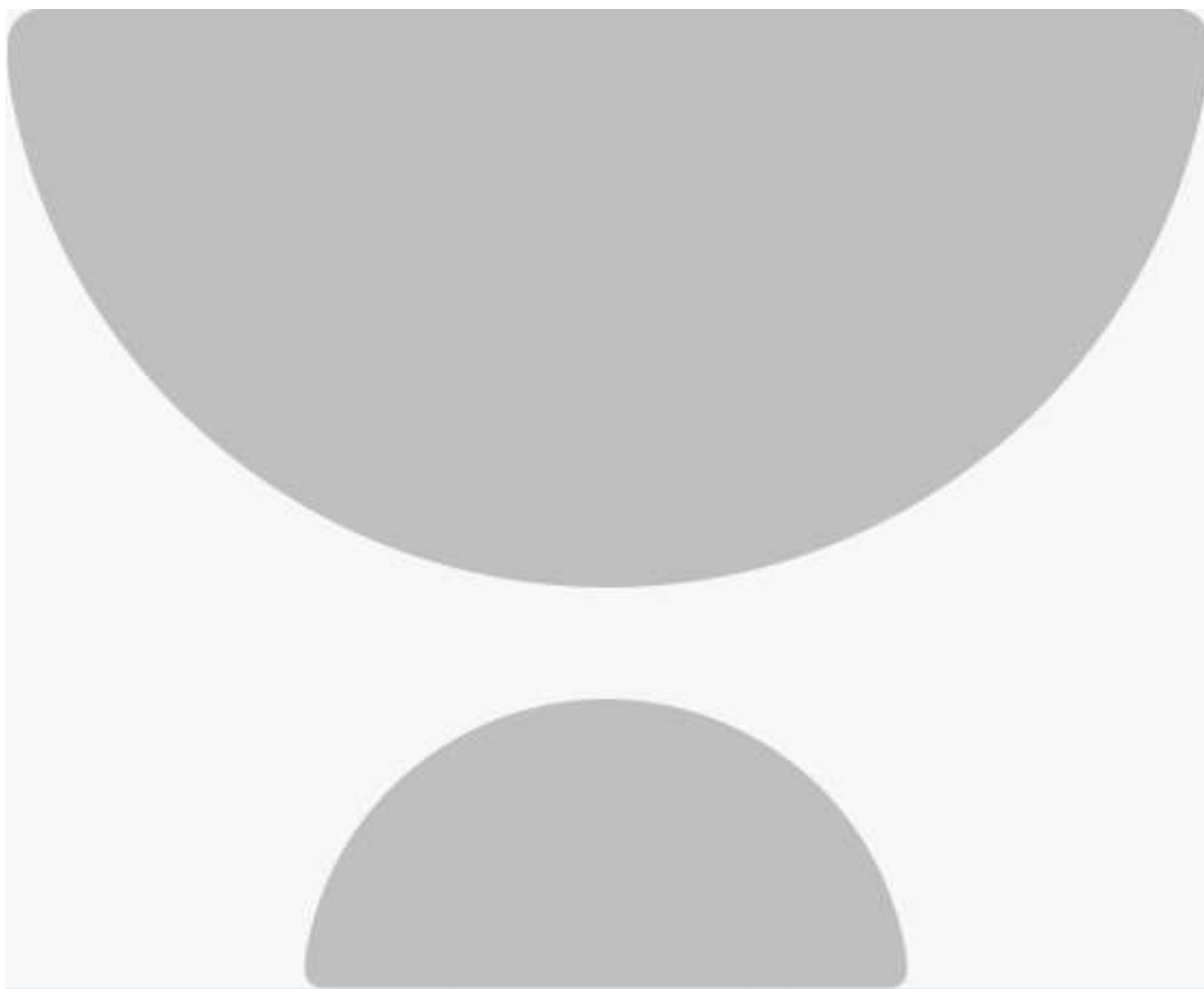
6. Which of the following compound is most reactive in electrophilic aromatic substitution?



Sol. (4)



+ R effect of -OH group enhances the reactivity in aromatic electrophilic substitution reaction.



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Alkyl halide & Aryl halide

Concept: Substitution reaction

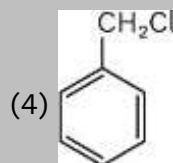
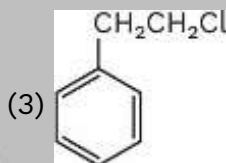
Sub concept: S_N1 reaction

Concept field: S_N1 reaction

7. Which of the following will NOT undergo S_N1 reaction with OH^- ?

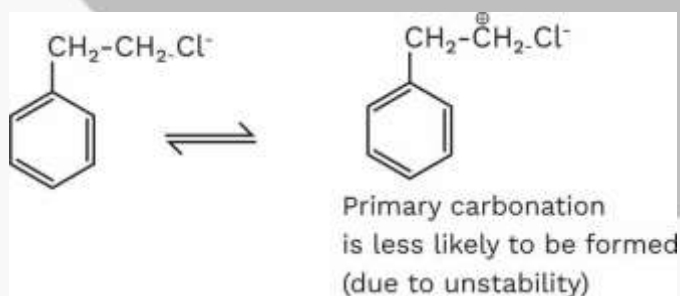
(1) $CH_2 = CH-CH_2Cl$

(2) $(CH_3)_3 CCl$



Sol. (3)

Reactivity in case of S_N1 reaction depends upon formation of carbocation.



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Chemistry in everyday life

Concept: Antibiotic

Sub concept: Broad spectrum

Concept field: Broad spectrum

8. Which of the following is not true about chloramphenicol ?

(1) It inhibits the growth of only grampositive bacteria.

(2) It is a broad spectrum antibiotic.

(3) It is not bactericidal.

(4) It is bacteriostatic.

Sol. (1)

Chloramphenicol is a broad-spectrum antibiotic which can inhibit the growth of gram positive bacteria and gram negative bacteria.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Polymer

Concept: Thermosetting Polymer

Sub concept: Condensation Reaction of Phenol + Formaldehyde

Concept field: Condensation Reaction of Phenol + Formaldehyde

9. Which of the following statement is correct about Bakelite?

- (1) It is a cross linked polymer. (2) It is an addition polymer.
(3) It is a branched chain polymer. (4) It is a linear polymer.

Sol. (1)

Bakelite is an example of cross-linked polymer.

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 sec.

Topic: Thermodynamics

Concept: Thermodynamics Equation

Sub concept: Spontaneous Process

Concept field: Spontaneous Process

10. If for a certain reaction $\Delta_r H$ is 30 kJ mol^{-1} at 450 K the value of $\Delta_r S$ (in $\text{JK}^{-1} \text{ mol}^{-1}$) for which the same reaction will be spontaneous at the same temperature is

- (1) 70 (2) -33
(3) 33 (4) -70

Sol. (1)

$$\Delta G = \Delta H - T \Delta S$$

For spontaneous, $\Delta G < 0$

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 sec.

Topic: Periodic Table

Concept: Classification of Elements

Sub concept: Electronic Configuration

Concept field: Vacant Subshell

11. Match the element in column I with that in Column II.

Column-I

- (a) Copper
- (b) Fluorine
- (c) Silicon
- (d) Cerium

Column-II

- (i) Non-Metal
- (ii) Transition metal
- (iii) Lanthanoid
- (iv) Metalloid

Identify the correct match :

- (1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Sol. (2)

Copper - Transition metal

Fluorine - Non metal

Silicon - Metalloids

Cerium - Lanthanoid

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Hydrocarbon

Concept: Substitution reaction

Sub concept: Free radical substitution reaction

Concept field: Free radical substitution reaction

12. Which of the following is a free radical substitution reaction?

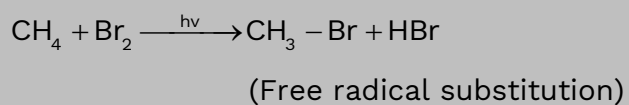
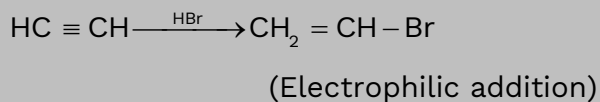
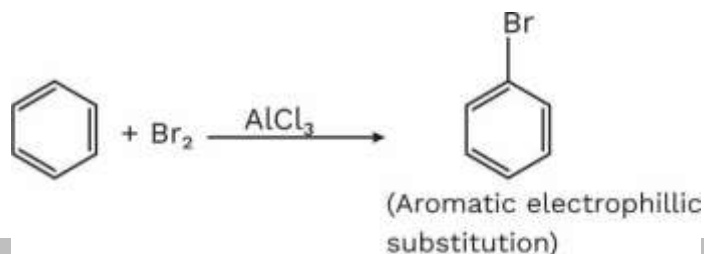
(1) Benzene with $\text{Br}_2/\text{AlCl}_3$

(2) Acetylene with HBr

(3) Methane with $\text{Br}_2/h\nu$

(4) Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{COO})_2$

Sol. (3)



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Carbohydrates

Concept: Concentrated sulphuric acid

Sub concept: Dehydration

Concept field: Dehydration

13. The reaction of concentrated sulphuric acid with carbohydrates ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is an example of

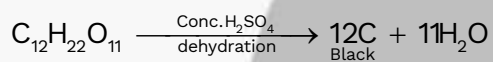
(1) Dehydration

(2) Oxidation

(3) Reduction

(4) Sulphonation

Sol. (1)



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

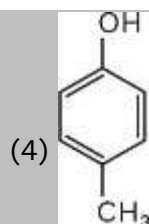
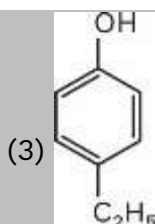
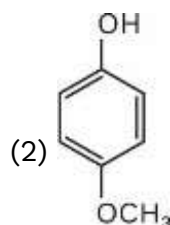
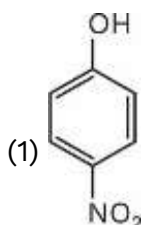
Topic: Phenol

Concept: Application of M, H, I effect

Sub concept: Acidic strength

Concept field: Acidic strength

14. Which of the following substituted phenols is the strongest acid ?



Sol. (1)

-NO₂ group is electron withdrawing group. Which increases the acidic strength of phenol.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Chemical Bonding

Concept: VSEPR Theory

Sub concept: Hybridization

Concept field: Geometry

15. Match the compounds of Xe in column I with the molecular structure in column II.

Column-I

Column-II

(a) XeF₂

(i) Square planar

(b) XeF₄

(ii) Linear

(c) XeO₃

(iii) Square pyramidal

(d) XeOF₄

(iv) Pyramidal

(1) (a)-(ii) (b)-(i) (c)-(iii) (d) - (iv)

(2) (a)-(ii) (b)-(iv) (c)-(iii) (d) - (i)

(3) (a)-(ii) (b)-(iii) (c)-(i) (d) - (iv)

(4) (a)-(ii) (b)-(i) (c)-(iv) (d) - (iii)

Sol. (4)

Unacademy Select

[34]

$\text{XeF}_2 \rightarrow \text{Linear}$

$\text{XeF}_4 \rightarrow \text{Square planar}$

$\text{XeO}_3 \rightarrow \text{Pyramidal}$

$\text{XeOF}_4 \rightarrow \text{Square pyramidal}$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Chemical Kinetics

Concept: Half-life of a Reaction

Sub concept: Calculation of Half-life of a Reaction

Concept field: Calculation of Half-life of a Reaction

16. The half-life for a zero-order reaction having 0.02 M initial concentration of reactant is 100s .The rate constant (in $\text{mol L}^{-1}\text{s}^{-1}$) for the reaction is

(1) 1.0×10^{-4}

(2) 2.0×10^{-4}

(3) 2.0×10^{-3}

(4) 1.0×10^{-2}

Sol. (1)

$$\begin{aligned} (t_{1/2})_{\text{Zero}} &= \frac{[A]_0}{2K} \\ 100\text{s} &= \frac{0.02\text{M}}{2K} \\ K &= 1 \times 10^{-4} \text{ molL}^{-1}\text{s}^{-1} \end{aligned}$$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: d-Block

Concept: Radius

Sub concept: Oxidation State

Concept field: Oxidation State

17. Identify the incorrect statement from the following:

- (1) Zirconium and Hafnium have identical radii of 160pm and 159pm, respectively as a consequence of lanthanoid contraction.
- (2) Lanthanoids reveal only +3 oxidation state.
- (3) The lanthanoid ions other than the f^0 type and the f^{14} type are all paramagnetic.
- (4) The overall decrease in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.

Sol.

(2)

Lanthanoids shows general oxidation state +3 but some elements can shows +2 as well as +4 .

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: d-Block

Concept: Transition Elements

Sub concept: Oxidation

Concept field: Variable Oxidation State

18. Match the following aspects with the respective metal.

Aspects	Metal
(a) The metal which reveals a Maximum number of oxidation states.	(i) Scandium
(b) The metal although placed in 3d Block is considered not as transition element.	(ii) Copper
(c) The metal which does not exhibit Variable oxidation states.	(iii) Manganese
(d) The metal which in +1 oxidation state In aqueous solution undergoes disproportionation.	(iv) Zinc

Select the correct option :

- (1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

Sol. (2)

In the given options

The metal which reveals a maximum number of oxidation state $\rightarrow$ Mn

The metal although placed in 3d block is considered not as a transition element is $\rightarrow$ Zn

The metal which does not exhibit variable oxidation state is $\rightarrow$ Sc (only +3)

The metal which in +1 oxidation state in aqueous solution undergoes disproportionation is $\rightarrow$ Cu

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Liquid Solution

Concept: Raoult's Law

Sub concept: Vapour Pressure

Concept field: Vapour Pressure

19. If 8g of a non-electrolyte solute is dissolved in 114g of n -octane to reduce its vapour pressure to 80%, the molar mass (in gmol^{-1}) of the solute is [Given that molar mass of n -octane is 114 gmol^{-1}]

(1) 40

(2) 60

(3) 80

(2) 20

Sol. (1)

Assuming dilute solution,

$$\frac{P_o - P_s}{P_s} \sim \frac{P_o - P_s}{P_o} = \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

$$\text{Let } P_o = 100, \text{ V.P. reduced to } 80\%, \therefore P_s = 80, \frac{100 - 80}{100} = \frac{8/m}{114/114}$$

$$m = 40$$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Coordination Compounds

Concept: Hybridization

Sub concept: Hybridization

Concept field: Coordination Number

20. Match the coordination number and type of hybridisation with distribution of hybrid orbitals in space on Valence bond theory.

Coordination number &

Type of hybridisation

(a) 4, sp^3

(b) 4, dsp^2

(c) 5, sp^3d

(d) 6, $d^2 sp^3$

Distribution of hybrid orbitals

in space

(i) trigonal bipyramidal

(ii) octahedral

(iii) tetrahedral

(iv) square planar

Select the correct option :

(1) (a)-(ii) (b)-(iii) (c)-(iv) (d)-(i)

(2) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)

(3) (a)-(iv) (b)-(i) (c)-(ii) (d)-(iii)

(4) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)

Sol. (2)

sp^3 – tetrahedral,

dsp^2 – square planar

sp^3d – trigonal bipyramidal,

d^2sp^3 -octahedral

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Atomic Structure

Concept: Nodes

Sub concept: Angular and Radial Nodes

Concept field: Angular and Radial Nodes

- 21.** The number of angular nodes and radial nodes in 3s orbital are
- | | |
|---------------------------|---------------------------|
| (1) 0 and 2, respectively | (2) 1 and 0, respectively |
| (3) 3 and 0, respectively | (4) 0 and 1, respectively |

Sol. (1)

No. of angular nodes = l

No. of Radial nodes = $n - l - 1$

For 3s ; $n = 3$ and $l = 0$

$\therefore$ No. of angular nodes = 0

$\therefore$ No. of radial nodes = 2

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: s-Block

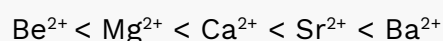
Concept: Properties of s-Block Elements

Sub concept: Chemical Properties

Concept field: Hydration Enthalpy

- 22.** Identify the correct statement from the following.

(1) The order of hydration enthalpies of alkaline earth cations



(2) Lithium and Magnesium show some similarities in their physical properties as they are diagonally placed in periodic table.

(3) Lithium is softer among all alkali metals.

(4) Lithium chloride is deliquescent and crystallises as a hydrate, $\text{LiCl} \cdot \text{H}_2\text{O}$.

Sol. (2)

Li & Mg shows diagonal relationship that's why they show similarity in their physical properties.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Chemistry in Everyday Life

Concept: Vitamins

Sub concept: Deficiency of Vitamins

Unacademy Select

Concept field: Deficiency of Vitamins

- 23.** Deficiency of which vitamin causes osteomalacia ?
- | | |
|---------------|---------------|
| (1) Vitamin A | (2) Vitamin D |
| (3) Vitamin K | (4) Vitamin E |

Sol. (2)
Deficiency of vitamin D causes osteomalacia
(soft bones and joint pain in adults)

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Chemical Bonding

Concept: VSEPR

Sub concept: Hybridization

Concept field: Shape of Molecule

- 24.** Identify the wrongly matched pair.

Molecule	Shape or geometry of molecule
(1) PCl_5	Trigonal planar
(2) SF_6	Octahedral
(3) BeCl_2	Linear
(4) NH_3	Trigonal pyramidal

Sol. (1)
 PCl_5 - Trigonal bipyramidal

Question Type: NEET

Difficulty of question: Moderate

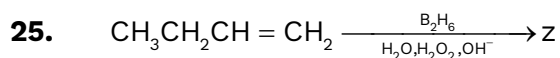
Expected time to solve: 25 sec.

Topic: Alcohol

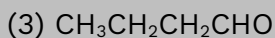
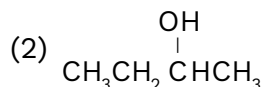
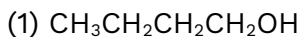
Concept: Preparation of alcohol

Sub concept: Hydroboration oxidation

Concept field: Hydroboration oxidation

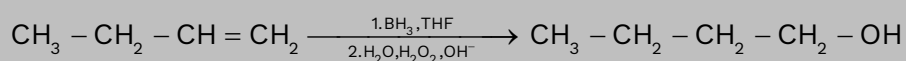


What is Z ?



Sol. (1)

Hydroboration - oxidation



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

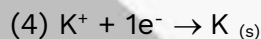
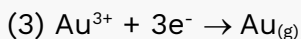
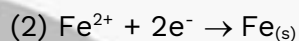
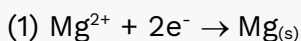
Topic: Electrochemistry

Concept: Electrode Potential

Sub concept: Standard Reduction Potential

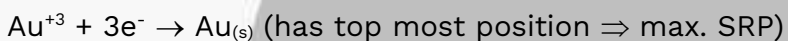
Concept field: Standard Reduction Potential

26. Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at 298K.



Sol. (3)

According to electrode potential series,



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Metallurgy

Concept: Extraction

Sub concept: Purification of Elements

Concept field: Purification of Elements

27. Match the elements in Column I with methods of purification in Column II.

Column I

Column II

(a) Boron

(i) Van Arkel method

(b) Tin

(ii) Mond's process

(c) Zirconium

(iii) Liquation

(d) Nickel

(iv) Zone refining

(1) (a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)

(2) (a)-(iv) (b)-(iii) (c)-(ii) (d)-(i)

(3) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)

(4) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)

Sol. (1)

B - Purified by Zone Refining

Sn - Liquation

Zr - Van arkel method

Ni - Mond's process

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Ionic Equilibrium

Concept: Salts Solution

Sub concept: Nature of Salts

Concept field: Nature of Salts

28. Which among the following salt solutions is basic in nature ?

(1) Ammonium chloride

(2) Ammonium sulphate

(3) Ammonium nitrate

(4) Sodium acetate

Sol. (4)

$\text{CH}_3\text{COONa} \rightarrow \text{Salt of } \text{CH}_3\text{COOH}(\text{WA}) + \text{NaOH}(\text{SB})$

Solution of CH_3COONa shows basic nature.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Surface Chemistry

Concept: Colloidal Particles

Sub concept: Sol Nature

Concept field: Sol Nature

29. In which of the sols, the colloidal particles are with negative charge ?

(1) TiO_2

(2) Haemoglobin

(3) Starch

(4) Hydrated Al_2O_3

Sol. (3)

Starch is example of negative sol.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

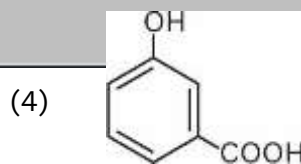
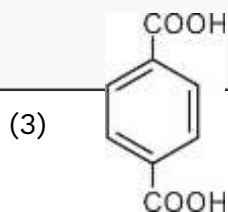
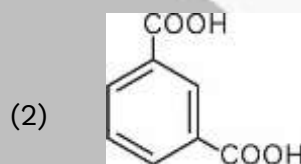
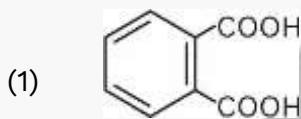
Topic: Amines

Concept: Heating

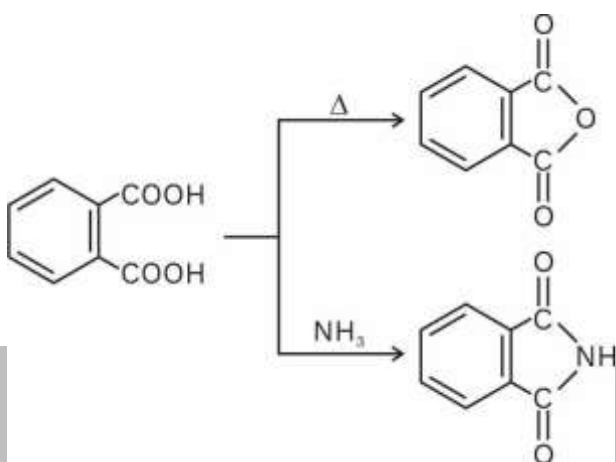
Sub concept: Heating with ammonia

Concept field: Heating with ammonia

30. Which of the following acid will form an (a) Anhydride on heating and (b) Acid imide on strong heating with ammonia ?



Sol. (1)



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Electrochemistry

Concept: Fuel Cell

Sub concept: Electrode Products

Concept field: Electrode Products

31. In a typical fuel cell, the reactants (R) and product (P) are

(1) $\text{R} = \text{H}_2(\text{g}), \text{O}_2(\text{g})$; ; $\text{P} = \text{H}_2\text{O}_2(\text{l})$

(2) $\text{R} = \text{H}_2(\text{g}), \text{O}_2(\text{g})$; $\text{P} = \text{H}_2\text{O}(\text{l})$

(3) $\text{R} = \text{H}_2(\text{g}), \text{O}_2(\text{g}), \text{Cl}_2(\text{g})$; $\text{P} = \text{HClO}_4(\text{aq})$

(4) $\text{R} = \text{H}_2(\text{g}), \text{N}_2(\text{g})$, $\text{P} = \text{NH}_3(\text{aq})$

Sol. (2)

In typical fuel cell

Reactants = H_2O_2

Products = H_2O

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 sec.

Topic: Chemical Kinetics

Concept: Collision Theory

Sub concept: Collision Frequency

Concept field: Collision Frequency

32. In collision theory of chemical reaction, Z_{AB} represents

(1) the fraction of molecules with energies greater than E_a

- (2) the collision frequency of reactants, A and B
- (3) steric factor
- (4) the fraction of molecules with energies equal to E_a

Sol. (2)

Z_{AB} = Collision frequency

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Biomolecules

Concept: Glucose

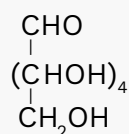
Sub concept: Aldohexose, reducing sugar

Concept field: Aldohexose, reducing sugar

33. Which of the following statement is not true about glucose ?

- (1) It is an aldohexose.
- (2) It contains five hydroxyl groups.
- (3) It is a reducing sugar.
- (4) It is an aldopentose.

Sol. (4)



Glucose

Glucose It is an aldohexose sugar.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

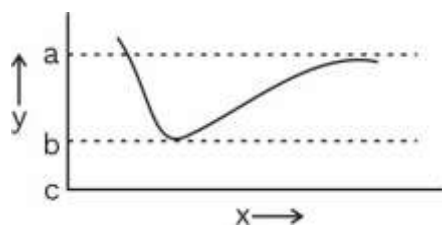
Topic: Chemical Bonding

Concept: Bond Energy

Sub concept: Bond Energy Analysis

Concept field: Bond Energy Analysis

34. The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below.



The bond energy of H_2 is :

(1) $(b-a)$

(2) $\frac{(c-a)}{2}$

(3) $\frac{(b-a)}{2}$

(4) $(c-a)$

Sol. (1)

As per the given curve bond energy is the amount of energy is released during the bond formation is i.e. = Final - Initial
 $= b - a$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Aldehyde & ketone

Concept: Reduction

Sub concept: Rosenmund reaction

Concept field : Rosenmund reaction

35. Identify compound (A) in the following reaction :



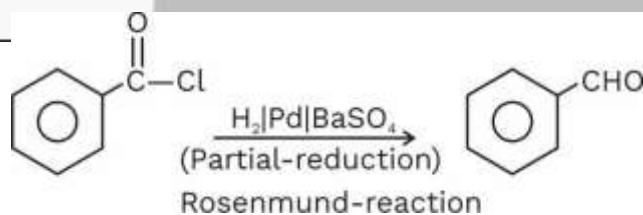
(1) Benzoyl chloride

(2) Toluene

(3) Acetophenone

(4) Benzoic acid

Sol. (1)



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

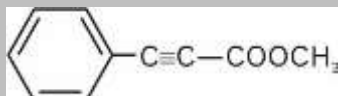
Topic: Basic of organic chemistry

Concept: Hybridization

Sub concept: Sigma & pie bond

Concept field: Sigma & pie bond

36. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound ?



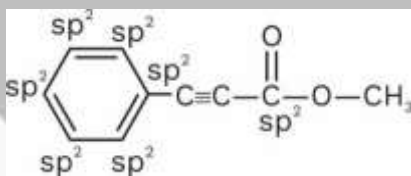
(1) 7, 5

(2) 8, 6

(3) 7, 6

(4) 8, 5

Sol. (3)



Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 sec.

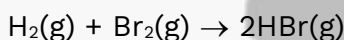
Topic: Thermo chemistry

Concept: Bond Energy

Sub concept: Calculation of Bond Energy of Reaction

Concept field: Calculation of Bond Energy of Reaction

37. At standard conditions, if the change in the enthalpy for the following reaction is -109 kJ mol^{-1}



Given that bond energy of H_2 and Br_2 is 435 kJ mol^{-1} and 192 kJ mol^{-1} , respectively, what is the bond energy (in kJ mol^{-1}) of HBr ?

(1) 368

(2) 736

(3) 518

(4) 259

Sol. (1)

$$\Delta H = [(\text{B.E})_{\text{Reactants}} - (\text{B.E})_{\text{Products}}]$$

$$-109 = [B.E_{(H-H)} + B.E_{(Br-Br)}] - [2 \times B.E_{(H-Br)}]$$

$$B.E_{(H-Br)} = 368 \text{ KJ/mol}$$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 35 sec.

Topic: States of Matter

Concept: Boyle's Law

Sub concept: Application of Boyle's Law

Concept field: Application of Boyle's Law

38. The minimum pressure required to compress 3 600dm of a gas at 1 bar to 3 150dm at 40 C° is

(1) 4.0 bar

(2) 0.2 bar

(3) 1.0 bar

(4) 2.5 bar

Sol.

(1)

By Boyle's law

$$P_1 V_1 = P_2 V_2$$

$$1 \text{ bar} \times 600 \text{ dm}^3 = P_2 \times 150 \text{ dm}^3$$

$$P_2 = 4 \text{ bar}$$

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: s-Block

Concept: Compounds of Calcium

Sub concept: Chemical Process in Cement

Concept field: Chemical Process in Cement

39. What is the role of gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ in setting of cement ? Identify the correct option from the following :

(1) to fasten the setting process

(2) to provide water molecules for hydration process

(3) to help to remove water molecules

(4) to slow down the setting process

Sol. (4)

The purpose of adding gypsum is only to slow down the process of setting of cement so that it gets sufficiently hardened.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: p-Block Element

Concept: Carbon Family

Sub concept: Nature of Compounds

Concept field: Oxidation

40. Which of the following oxide is amphoteric in nature?

(1) SnO_2

(2) SiO_2

(3) GeO_2

(4) CO_2

Sol. (1) SnO_2 – amphoteric

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 sec.

Topic: Redox Reaction

Concept: Chemical Reaction

Sub concept: Hydrolysis

Concept field: Hydrolysis

41. Which one of the following reactions does not come under hydrolysis type reaction?

(1) $\text{SiCl}_{4(l)} + 2\text{H}_2\text{O}_{(l)} \rightarrow \text{SiO}_{2(s)} + 4\text{HCl}_{(aq)}$

(2) $\text{Li}_2\text{N}_{(s)} + 3\text{H}_2\text{O}_{(l)} \rightarrow \text{NH}_{3(g)} + 3\text{LiOH}_{(aq)}$

(3) $2\text{F}_{2(g)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 4\text{HF}_{(aq)} + \text{O}_{2(g)}$

(4) $\text{P}_4\text{O}_{10(s)} + 6\text{H}_2\text{O}_{(l)} \rightarrow 4\text{H}_3\text{PO}_{4(aq)}$

Sol. (3)

$2\text{F}_{2(g)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 4\text{HF}_{(aq)} + \text{O}_{2(g)}$

It's a type of Redox reaction.

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 sec.

Topic: Solid State

Concept: Crystal Defects

Sub concept: Defects

Concept field: Defects

42. Which one of the following compounds shows both, Frenkel as well as Schottky defects ?

(1) AgBr

(2) AgI

(3) NaCl

(4) ZnS

Sol.

(1)

AgBr shows both Schottky and Frenkel defect

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 sec.

Topic: Mole Concepts

Concept: No. of Atoms

Sub concept: Moles

Concept field: Moles

43. One mole of carbon atomic weight 12g, the number of atoms in it is equal to, (Mass of carbon -12 is 1.9926×10^{-23} g)

(1) 1.2×10^{23}

(2) 6.022×10^{22}

(3) 12×10^{22}

(4) 6.022×10^{23}

Sol.

(4)

1 mole of carbon = 6.022×10^{23} atoms

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 15 sec.

Topic: Liquid Solution

Concept: Isotonic Solutions

Sub concept: Colligative Properties

Concept field: Colligative Properties

44. Isotonic solutions have same

(1) vapour pressure

(2) freezing temperature

(3) osmotic pressure

(4) boiling temperature

Sol. (3)

Isotonic solutions have same osmotic pressure.

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 sec.

Topic: Ionic Equilibrium

Concept: Solubility Product

Sub concept: Molarity

Concept field: Molarity

45. The solubility product for a salt of the type AB is 4×10^{-8} . What is the molarity of its standard solution ?

(1) 2×10^{-4} mol/L

(2) 16×10^{-16} mol/L

(3) 2×10^{-16} mol/L

(4) 4×10^{-4} mol/L

Sol. (1)

K_{sp} for AB = $s^2 = 4 \times 10^{-8}$

$\therefore$ Molarity of solution = solubility = $\sqrt{K_{sp}}$

$= \sqrt{4 \times 10^{-8}}$

$= 2 \times 10^{-4}$ mol/L

Biology NEET 2020

Class 11th

Question Type: NEET

Difficulty of question: Easy

Unacademy Select

[51]

Expected time to solve: 15 secs

Topic: Morphology of Flowering Plant

Concept: Fruit

Sub-concept: Types of Fruit

Concept field: Types of Fruit

1. In some plants thalamus contributes to fruit formation. Such fruits are termed as
- | | |
|------------------|---------------------------|
| (1) False fruits | (2) Aggregate fruits |
| (3) True fruits | (4) Parthenocarpic fruits |

Answer (1)

Sol. False fruits: Fruits that arise from floral parts other than ovary.
e.g., Apple, Pineapple, Banana.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 secs

Topic: Biotechnology: Principles and Processes

Concept: Tools of Recombinant DNA

Sub-concept: Restriction Enzymes

Concept field: Restriction Site

2. First discovered restriction endonuclease that always cuts the DNA molecule at a particular point by recognising a specific sequence of six base pairs is
- | | |
|---------------------------------|-------------------------|
| (1) EcoR1 | (2) Adenosine deaminase |
| (3) Thermostable DNA polymerase | (4) Hind II |

Answer (4)

Sol. First discovered restriction endonucleases -Hind II, always cut DNA molecules at a particular point by recognizing a specific sequence of six base pairs.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 25 secs

Topic: Ecosystem

Concept: Trophic Level

Sub-concept: Ecological Pyramids

Concept field: Ecological Pyramids

3. Which of the following statements is **incorrect**?
- | |
|--|
| (1) Biomass decreases from first to fourth trophic level |
| (2) Energy content gradually increases from first to fourth trophic level |
| (3) Number of individuals decreases from first trophic level to fourth trophic level |
| (4) Energy content gradually decreases from first to fourth trophic level |

Answer (2)

Sol. According to ten percent law by Lindemann- During the transfer of organic food energy from one trophic level to the next, only 10% of transferred energy is stored.

Class 12th

Question Type: NEET/AIIMS

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Molecular Basis of Inheritance

Concept: Genetic Materials

Sub-concept: History of Genetic Materials

Concept field: DNA

4. The term 'Nuclein' for the genetic material was used by
(1) Franklin (2) Miescher (3) Chargaff (4) Mendel

Answer (2)

Sol. Johann Friedrich Miescher (1869), discovered a new substance in nucleus of living cells, which he called as **nuclein**.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 10 secs

Topic: Principles of Inheritance and Variation

Concept: Inheritance of Two Genes

Sub-concept: History of Chromosomal Inheritance

Concept field: Chromosomal Theory of Inheritance

5. Chromosomal theory of inheritance was proposed by
(1) Sutton and Boveri (2) Bateson and Punnet
(3) T. H. Morgan (4) Watson and Crick

Answer (1)

Sol. Chromosomal theory of inheritance was proposed by-Sutton and Boveri.

Class 11th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 10 secs

Topic: Plant Kingdom

Concept: Algae

Sub-concept: Major Pigments

Concept field: Rhodophyceae

6. Phycoerythrin is the major pigment in
(1) Red algae (2) Blue green algae
(3) Green algae (4) Brown algae

Answer (1)

Sol. Major pigments of Rhodophyceae are-chlorophyll a, d and phycoerythrin.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Biomolecules

Concept: Amino Acids

Sub-concept: Chemical Composition

Concept field: Chemical Composition

7. Identify the statement which is **incorrect**.

-
- (1) Sulphur is an integral part of cysteine.
 - (2) Glycine is an example of lipids.
 - (3) Lecithin contains phosphorus atom in its structure.
 - (4) Tyrosine possesses aromatic ring in its structure.

Answer (2)

Sol. Glycerol, Phospholipids and cholesterol are example of lipids.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Plant Kingdom

Concept: Divisions in Plant Kingdom

Sub-concept: Gymnosperms

Concept field: Gametophytes

8. Which of the following statements is incorrect about gymnosperms?
- (1) They are heterosporous
 - (2) Male and female gametophytes are free living
 - (3) Most of them have narrow leaves with thick cuticle
 - (4) Their seeds are not covered

Answer (2)

Sol. Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophytes do not have an independent free living existence.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Environmental Issues

Concept: Water Pollution and its Control

Sub-concept: Aquatic Weed

Concept field: Terror of Bengal

9. A species which was introduced for ornamentation but has become a trouble-some weed in India.
- (1) *Parthenium hysterophorus*
 - (2) *Eichhornia crassipes*
 - (3) *Prosopis juliflora*
 - (4) *Trapa spinosa*

Answer (2)

Sol. *Eichhornia crassipes* (water hyacinth) is the world's most problematic aquatic weed, also called 'Terror of Bengal'.

They grow abundantly in eutrophic water bodies, and lead to an imbalance in the ecosystem dynamics of the water body.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Morphology of Flowering Plants

Concept: Flower

Sub-concept: Types of Flower

Concept field: Hypogynous Flower

10. Correct position of floral parts over thalamus in mustard plant is

Unacademy Select

[54]

-
- (1) Gynoecium occupies the highest position, while the other parts are situated below it.
 - (2) Margin of the thalamus grows upward, enclosing the ovary completely, and other parts arise below the ovary.
 - (3) Gynoecium is present in the centre and other parts cover it partially.
 - (4) Gynoecium is situated in the centre, and other parts of the flower are located at the rim of the thalamus, at the same level.

Answer (1)

Sol. In mustard, the flower is hypogynous i.e., the gynoecium occupies the highest position while the other parts are situated below it.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Biotechnology – Principles and Processes

Concept: Cloning Vectors

Sub-concept: Cloning Site

Concept field: Antibiotic Resistance Genes

11. In Recombinant DNA technology antibiotics are used
- (1) to keep medium bacteria-free
 - (2) to detect alien DNA
 - (3) to impart disease-resistance to the host plant
 - (4) as selectable markers

Answer (4)

Sol. In recombinant DNA technology, antibiotics are used as selectable markers.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Biodiversity and Conservation

Concept: Biodiversity

Sub-concept: Pattern of Biodiversity

Concept field: Species-Area Relationship

12. According to Alexander von Humboldt
- (1) Species richness decreases with increasing area of exploration
 - (2) Species richness increases with increasing area, but only up to limit
 - (3) There is no relationship between species richness and area explored.
 - (4) Species richness goes on increasing with increasing area of exploration

Answer (2)

Sol. Alexander von Humboldt while exploring south American jungles, observed that within a region, species richness increases with increasing explored area, but only up to a limit.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 secs

Topic: Sexual Reproduction in Flowering Plants

Concept: Agents of Pollination

Sub-concept: Agents of Pollination

Concept field: Wind Pollination

13. Which of the following is **incorrect** for wind-pollinated plants?

- (1) Well exposed stamens and stigma
- (2) Many ovules in each ovary
- (3) Flowers are small and not brightly coloured
- (4) Pollen grains are light and non-sticky

Answer (2)

Sol. In wind pollinated plants (Anemophily), wind is the pollinating agent. Hence, the flower must have light weight, non sticky pollens, well exposed stigma and stamens and long and slender filaments.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Morphology of Flowering Plants

Concept: Semi-technical Description of Flowering Plants

Sub-concept: Floral Formulae

Concept field: Floral Formulae

14. Which of the following is the correct floral formula of Liliaceae?

- (1) $\% \overline{\text{C}}_{1+2+(2)} \text{A}_{(9)+1} \underline{\text{G}}_1$
- (2) $\oplus \overline{\text{C}}_{(5)} \text{A}_5 \underline{\text{G}}_{(2)}$
- (3) $\text{Br} \oplus \overline{\text{P}}_{(3+3)} \text{A}_{3+3} \text{G}_{(3)}$
- (4) $\oplus \overline{\text{C}}_{(5)} \text{A}_5 \underline{\text{G}}_{(2)}$

Answer (3)

Sol. Floral formulae of Liliaceae is: $\text{Br} \oplus \overline{\text{P}}_{(3+3)} \text{A}_{3+3} \text{G}_{(3)}$

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Mineral Nutrition

Concept: Biological Nitrogen Fixation

Sub-concept: Nodule Formation

Concept field: Nodule Formation in Soyabean

15. In Glycine max, the product of biological nitrogen fixation is transported from the root nodules to other parts as

- (1) Ammonia
- (2) Glutamate
- (3) Nitrates
- (4) Ureides

Answer (4)

Sol. In soyabean, products of fixation are the ureides allantoin and allantoic acid.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 10 secs

Topic: Principles of Inheritance and Variation

Concept: Mendel's Law of Inheritance

Sub-concept: Mendel's Experiment

Concept field: Seven Pairs of Contrasting Traits

-
- 16.** The number of contrasting characters studied by Mendel for his experiment were
(1) 14 (2) 4 (3) 2 (4) 7

Answer (4)

Sol. Mendel used 7 number of contrasting characters in his experiment, namely- flower colour, seed shape, seed colour, pod shape, pod colour, stem height and flower position.



Class 11th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 15 secs**Topic:** Cell Cycle and Cell Division**Concept:** Phases of Cell Cycle**Sub-concept:** M Phase**Concept field:** Metaphase

17. Attachment of spindle fibres to kinetochores of chromosomes becomes evident in
(1) Anaphase (2) Telophase (3) Prophase (4) Metaphase

Answer (4)

Sol. During metaphase, spindle fibres attach to the kinetochores of chromosomes.

Class 12th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 25 secs**Topic:** Organisms and Populations**Concept:** Population**Sub-concept:** Population Interactions**Concept field:** Types of Interaction

18. Match the items in Column-I with those in Column-II.

Column I

- (a) Herbivores-Plants
- (b) Mycorrhiza-Plants
- (c) Sheep-Cattle
- (d) Orchid-Tree

Column II

- (i) Commensalism
- (ii) Mutualism
- (iii) Predation
- (iv) Competition

Select the correct option from following:

- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
- (3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

Answer (2)

- Sol.**
- The Relationship between herbivores and plants is Predation.
 - Mycorrhiza and plants are mutually symbiotic.
 - Mycorrhiza increases the absorption of nutrients (phosphorous) for plants and plants in turn provides organic nutrients to the fungi.
 - Sheep and cattle are herbivores and therefore, both are competitive for consuming the same resource.
 - The tree and epiphytic orchids are commensals, the orchid is unable to survive without the aid of the tree and trees get no harm.

Class 12th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 15 secs**Topic:** Reproduction in Organisms**Concept:** Types of Reproduction**Sub-concept:** Asexual Reproduction**Concept field:** Vegetative Propagules

19. Vegetative propagule in *Agave* is
(1) Rhizome (2) Bulbil (3) Offset (4) Eye

Answer (2)

Vegetative propagule in *Agave* is bulbil.

Class 11th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 25 secs**Topic:** Biomolecules.**Concept:** Biomacromolecules.**Sub-concept:** Types of Macromolecules**Concept field:** Examples of Macromolecules**20.** Match the following.

- | | |
|--------------------|---------------------|
| (a) Aquaporin | (i) Amide |
| (b) Asparagine | (ii) Polysaccharide |
| (c) Absciscic acid | (iii) Polypeptide |
| (d) Chitin | (iv) Carotenoids |

Select the correct option:

- | | |
|--|--|
| (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) | (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) |
| (3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii) | (4) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv) |

Answer (1)

- Sol.**
- Aquaporin in intestines are Vasoactive intestinal polypeptide (VIP).
 - Asparagine is an amide type amino acid.
 - Absciscic acid is one of the products of metabolic pathway of carotenoids.
 - Chitin is a polysaccharide made of N-AG.

Class 11th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 20 secs**Topic:** Mineral Nutrition**Concept:** Types of Nutrients: Macro and Micro**Sub-concept:** Role of Macro and Micronutrients**Concept field:** Role of Magnesium**21.** Which of the following elements helps in maintaining the structure of ribosomes?

- | | | | |
|---------------|----------|------------|----------------|
| (1) Magnesium | (2) Zinc | (3) Copper | (4) Molybdenum |
|---------------|----------|------------|----------------|

Answer (1)

Sol. Magnesium helps in maintaining the structure of ribosomes.

Class 11th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 15 secs**Topic:** Plant Growth and Development**Concept:** Plant Growth Regulators**Sub-concept:** Discovery of Plant Growth Regulators**Concept field:** Discovery of Cytokinin**22.** Who coined the term 'Kinetin'?

- | | |
|----------------------|--------------|
| (1) Skoog and Miller | (2) Darwin |
| (3) Went | (4) Kurosawa |

Answer (1)

Sol. Skoog and Miller identified the active substance responsible for proliferation and called it kinetin.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Biodiversity and Conservation

Concept: Ways of Conservation

Sub-concept: *In-situ* and *Ex-situ*

Concept field: Examples

23. In the following in each set a conservation approach and an example of method of conservation are given

- (a) *In situ* conservation - Biosphere Reserve
- (b) *Ex situ* conservation - Sacred groves
- (c) *In situ* conservation - Seed bank
- (d) *Ex situ* conservation - Cryopreservation

Select the option with correct match of approach and method:

- (1) (a) and (c) (2) (a) and (d) (3) (b) and (d) (4) (a) and (b)

Answer (2)

- Sol.**
- *In situ* conservation-protecting an endangered plants or animal in its natural habitat.
e.g. Biosphere reserves, national parks.
 - *Ex-situ* conservation-protecting techniques which involves the transfer of a target species away from its native place.
e.g. – Zoological garden, Botanical garden, Seed bank, Cryopreservation.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Evolution

Concept: Evolutionary Theories

Sub-concept: Evidence for Evolution

Concept field: Embryological Support for Evolution

24. Embryological support for evolution was proposed by

- (1) Ernst Haeckel
- (2) Karl Ernst von Baer
- (3) Charles Darwin
- (4) Alfred Wallace

Answer (1)

Sol. Embryological support for evolution was proposed by-Ernst Haeckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Photosynthesis

Concept: Light and Dark Reactions

Sub-concept: Non-cyclic Photophosphorylation

Concept field: Z Scheme of Light Reaction

Unacademy Select

25. During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS II, what is the source which replaces these electrons?
(1) Oxygen (2) Water (3) Carbon dioxide (4) Light

Answer (2)

Sol. Water is the source of electron in non-cyclic photophosphorylation.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 10 secs

Topic: Cell Cycle and Cell Division

Concept: Cell Cycle

Sub-concept: Phases of Cell Cycle

Concept field: Sequence of Cell Cycle Phases

26. In a mitotic cycle, the correct sequence of phases is

(1) S, G₁, G₂, M (2) G₁, S, G, M (3) M, G₁, G₂, S (4) G, G₂, S, M

Answer (2)

Sol. Correct sequence of phases in cell cycle is G₁ → S → G₂ → M.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Cell: The Unit of Life

Concept: Prokaryotic Cells

Sub-concept: Reserve Material

Concept field: Inclusion Bodies of Bacteria

27. Inclusion bodies of blue- green, purple and green photosynthetic bacteria are

(1) Contractile vacuoles (2) Gas vacuoles
(3) Centrioles (4) Microtubules

Answer (2)

Sol. Gas vacuoles are the reserve materials in blue green, purple and green photosynthetic bacteria.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Anatomy of Flowering Plants

Concept: Anatomy of Leaf

Sub-concept: Monocot Leaf

Concept field: Bulliform Cells

28. Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are

(1) Lenticels (2) Guard cells
(3) Bundle sheath cells (4) Bulliform cells

Answer (4)

Sol. Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are Bulliform cells.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Cell : The Unit of Life

Concept: Eukaryotic Nucleus

Sub-concept: Functions of Components of Nucleus

Concept field: Function of Nucleolus

29. The biosynthesis of ribosomal RNA occurs in

- (1) Ribosomes (2) Golgi apparatus (3) Microbodies (4) Nucleolus

Answer (4)

Sol. Biosynthesis of rRNA occurs in Nucleolus.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Biological Classification

Concept: Five Kingdom Classification

Sub-concept: Kingdom Monera

Concept field: Cyanobacteria

30. Which of the following is incorrect about Cyanobacteria?

- (1) They are photoautotrophs
(2) They lack heterocysts
(3) They often form blooms in polluted water bodies
(4) They have chlorophyll- a, similar to green plants

Answer (2)

Sol. Cyanobacteria (blue-green algae) have chlorophyll-a and thus are photoautotrophs. They often forms algal blooms in polluted water bodies.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Anatomy of Flowering Plants

Concept: Secondary Growth in Stem

Sub-concept: Cork Cambium

Concept field: Cork Cambium

31. Which of the following statements about cork cambium is **incorrect**?

- (1) It forms secondary cortex on its outside
(2) It forms a part of periderm
(3) It is responsible for the formation of lenticels
(4) It is a couple of layers thick

Answer (1)

Sol. Cork cambium forms a part of periderm, forms lenticels and is two layered thick.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Transport in Plants

Concept: Uptake and Transport of Mineral Nutrients

Unacademy Select

Sub-concept: Phloem Transport

Concept field: Pressure Flow Hypothesis

32. Select the **incorrect** statement.

- (1) Transport of molecules in phloem can be bidirectional.
- (2) Movement of minerals in xylem is unidirectional.
- (3) Unloading of sucrose at sink does not involve the utilization of ATP.
- (4) Elements most easily mobilized in plants from one region to another are: phosphorus, sulphur, nitrogen and potassium.

Answer (3)

Sol. Transport in phloem is bidirectional, whereas in xylem it is unidirectional. Elements like P, S, N and K are easily mobilized from one region to another.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 15 secs

Topic: Environmental Issues

Concept: Pollution

Sub-concept: Air pollution and its Control

Concept field: Air Pollutants

33. Air (Prevention and Control of Pollution) Act was amended in 1987 to include among pollutants

- (1) Vehicular exhaust
- (2) Allergy causing pollen
- (3) Noise
- (4) Particulates of size 2.5 micrometre or below

Answer (3)

Sol. Noise is also included among air pollutants after the amendment (1987) done in Air prevention and control of Pollution Act.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Plant Growth and Development

Concept: Plant Growth Inhibitors

Sub-concept: Seed Dormancy and Inhibitors

Concept field: Seed Dormancy and Inhibitors

34. Inhibitory substances in dormant seeds cannot be removed by subjecting seeds to

- (1) Gibberellic acid (2) Nitrate (3) Ascorbic acid (4) Chilling conditions

Answer (3)

Sol. Inhibitory substances in dormant seeds can be removed by subjecting seeds to Gibberellic acid, Nitrate and chilling conditions.

Class 12th

Question Type: NEET

Difficulty of question: Moderate/Difficult

Expected time to solve: 25 secs

Topic: Biotechnology and its Applications

Concept: Biotechnological Applications in Medicine

Sub-concept: Molecular Diagnosis

Concept field: Techniques for Molecular Diagnosis

35. Match the following techniques or instruments with their usage.

- | | |
|---------------------|---|
| (a) Bioreactor | (i) Separation of DNA fragments |
| (b) Electrophoresis | (ii) Production of large quantities of products |
| (c) PCR | (iii) Detection of pathogen, based on antigen – antibody reaction |
| (d) ELISA | (iv) Amplification of nucleic acids |

Select the correct option from following:

- | | |
|--|--|
| (1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) | (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii) |
| (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) | (4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv) |

Answer (2)

- Sol.**
- Large quantities of products are produced using Bioreactor.
 - DNA fragments are separated using electrophoresis.
 - Amplification of nucleic acid are done by PCR.
 - ELISA is used to detect pathogen.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 25 secs

Topic: Photosynthesis in Higher Plants

Concept: C₄ and C₃ Pathway

Sub-concept: Enzymes

Concept field: Output of C₃ and C₄ Pathways

36. Which of the following statements is **incorrect**?

- (1) RuBisCO is a bifunctional enzyme
- (2) In C₄ plants, the site of RuBisCO activity is mesophyll cell
- (3) The substrate molecule for RuBisCO activity is a 5-carbon compound
- (4) RuBisCO action requires ATP and NADPH

Answer (2)

Sol. In C₄ plants, RuBisCo is present in Bundle sheath cells.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Ecosystem

Concept: Nutrient Cycling

Sub-concept: Sedimentary Cycles

Concept field: Phosphorus Cycle

37. Which of the following statements is incorrect regarding the phosphorus cycle?

- (1) Phosphates are the major form of phosphorus reservoir
- (2) Phosphorus solubilising bacteria facilitate the release of phosphorus from organic remains
- (3) There is appreciable respiratory release of phosphorus into atmosphere
- (4) It is sedimentary cycle

Answer (3)

Sol. The natural reservoir of phosphorous is rock, which contains phosphorous in the form of phosphate. Thus, there is no respiratory release of phosphorus into the atmosphere.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 secs

Topic: Evolution

Concept: Origin of Life

Sub-concept: Formation of Earth

Concept field: Appearance of Life

38. After about how many years of formation of earth, life appeared on this planet?

- | | |
|-----------------------|----------------------|
| (1) 500 billion years | (2) 50 million years |
| (3) 500 million years | (4) 50 billion years |

Answer (3)

Sol. Life appeared on this planet 500 million years ago , after the formation of life.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Biotechnology: Principles and Process

Concept: Restriction Enzymes

Sub-concept: Separation and Isolation of DNA Fragments

Concept field: Separation and Isolation of DNA Fragments

39. In a mixture, DNA fragments are separated by

- | | |
|----------------------------|-------------------------------|
| (1) Bioprocess engineering | (2) Restriction digestion |
| (3) Electrophoresis | (4) Polymerase chain reaction |

Answer (3)

Sol. DNA fragments are separated by the process of Electrophoresis.

Class 11th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 30 secs

Topic: Morphology in Flowering Plants

Concept: Fruit

Sub-concept: Formation of a Fruit

Concept field: Types of Fruit

40. Identify the correct features of Mango and Coconut fruits.

- (i) In both fruit is a drupe
- (ii) Endocarp is edible in both
- (iii) Mesocarp in Coconut is fibrous, and in Mango it is fleshy
- (iv) In both, fruit develops from monocarpellary ovary

Select the correct option from below:

- | | |
|------------------------------|------------------------------|
| (1) (i), (iii) and (iv) only | (2) (i), (ii) and (iii) only |
| (3) (i) and (iv) only | (4) (i) and (ii) only |

Answer (1)

Sol. Endocarp in mango is not edible.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 15 secs

Topic: Organisms and Populations

Concept: Population Growth

Sub-concept: Factors Affecting Population Density

Concept field: Immigration

41. The impact of immigration on population density is
- | | |
|-----------------------------|--------------------------------|
| (1) Negative | (2) Both positive and negative |
| (3) Neutralized by natality | (4) Positive |

Answer (4)

Sol. Immigration adds more number of individuals to the existing population density. Thus, its impact is positive.

Class 11th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 20 secs

Topic: Plant Kingdom

Concept: Divisions in Plants

Sub-concept: Gametophytes in Plants

Concept field: Gametophyte in Angiosperms

42. Male and female gametophytes do not have an independent free living existence in :-
- | | | | |
|-------------------|-----------|-----------------|----------------|
| (1) Pteridophytes | (2) Algae | (3) Angiosperms | (4) Bryophytes |
|-------------------|-----------|-----------------|----------------|

Answer (3)

Sol. In angiosperms, the male and female gametophytes do not have an independent free living existence (enclosed in a flower)

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Plants Growth and Hormones

Concept: Phytohormones

Sub-concept: Functions of Phytohormones

Concept field: Functions of Phytohormones

43. Match the following concerning the activity/function and the phytohormone involved.
- | | |
|--------------------|----------------------|
| (a) Fruit ripener | (i) Absciscic acid |
| (b) Herbicide | (ii) GA ₃ |
| (c) Bolting agent | (iii) 2, 4-D |
| (d) Stress hormone | (iv) Ethephon |
- Select the correct option from following.
- | | |
|--|--|
| (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) | (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) |
| (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) | (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii) |

Answer (3)

Sol.

- Ethephon promotes fruit ripening.
- 2, 4-D is a herbicide.
- GA₃ (gibberellic acid) causes bolting effect.
- ABA (Absciscic acid) is also called as a stress hormone.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Respiration in Plants

Concept: Aerobic Respiration

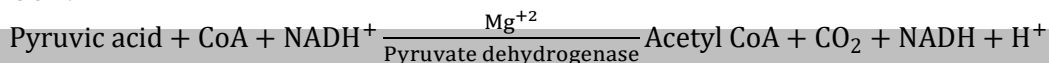
Sub-concept: Conversion of Pyruvic Acid to Acetyl CoA

Concept field: Role of Pyruvate Dehydrogenase and Mg^{+2}

44. Pyruvate dehydrogenase activity during aerobic respiration requires
(1) Calcium (2) Iron (3) Cobalt (4) Magnesium

Answer (4)

Sol. Pyruvate dehydrogenase requires magnesium for conversion of pyruvate to Acetyl CoA.



Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Ecosystem

Concept: Energy Flow

Sub-concept: Detritus Food Chain

Concept field: Detritus Food Chain

45. The rate of decomposition is faster in the ecosystem due to following factors except
(1) Detritus rich in sugars (2) Warm and moist environment
(3) Presence of aerobic soil microbes (4) Detritus richer in lignin and chitin

Answer (4)

Sol. Rate of decomposition is faster in ecosystem as detritus is rich in sugar, warm and moist and aerobic soil microbes are present to facilitate decomposition.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Microbes in Human Welfare

Concept: Microbes in Industrial Products

Sub-concept: Production of Citric Acid

Concept field: Production of Citric Acid

46. For the commercial and industrial production of Citric Acid, which of the following microbes is used?
(1) *Aspergillus niger* (2) *Lactobacillus sp.*
(3) *Saccharomyces cerevisiae* (4) *Clostridium butylicum*

Answer (1)

Sol. *Aspergillus niger* is used for the commercial and industrial production of citric acid.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Reproductive Health

Concept: Sexually Transmitted Diseases (STDs)

Sub-concept: Types of STDs

Concept field: Cure for STDs

47. Which of the following STDs are **not** curable?

-
- (1) Genital herpes, Hepatitis B, HIV infection
 - (2) Chlamydiosis, Syphilis, Genital warts
 - (3) HIV, Gonorrhoea, Trichomoniasis
 - (4) Gonorrhoea, Trichomoniasis, Hepatitis B

Answer (1)

Sol. STDs like herpes, Hepatitis B and HIV are not curable till date.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Biotechnology – Principles and Processes

Concept: Process of Recombinant DNA Technology

Sub-concept: Isolation of DNA

Concept field: Spooling Process

- 48.** Spooling is
- (1) Amplification of DNA
 - (2) Cutting of separated DNA bands from the agarose gel
 - (3) Transfer of separated DNA fragments to synthetic membranes
 - (4) Collection of isolated DNA

Answer (4)

Sol. Collection of isolated DNA in form of spool over a glass rod is called spooling.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Evolution

Concept: Evidence for Evolution

Sub-concept: Darwin's Theory

Concept field: Adaptive Radiation

- 49.** The phenomenon of evolution of different species in a given geographical area starting from a point and spreading to other habitats is called
- (1) Saltation
 - (2) Co-evolution
 - (3) Natural selection
 - (4) Adaptive radiation

Answer (4)

Sol. Adaptive radiation is evolution of different species in a given geographical area starting from a point and spreading to other habitat.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Principles of Inheritance and Variation

Concept: Pleiotropic

Sub-concept: Pleiotropic Genes

Concept field: Disease Caused by Mutation in Pleiotropic Genes

- 50.** The best example for pleiotropy is
- (1) Skin colour
 - (2) Phenylketonuria
 - (3) Colour Blindness
 - (4) ABO Blood group

Answer (2)

Sol. Phenylketonuria is one of the best examples of Pleiotropy.

Unacademy Select

[68]

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Structural Organisation in Animals

Concept: Anatomy

Sub-concept: Alimentary Canal of Cockroach

Concept field: Alimentary Canal of Cockroach

51. In cockroach, identify the parts of the foregut in correct sequence

- (1) Mouth → Oesophagus → Pharynx → Crop → Gizzard
- (2) Mouth → Crop → Pharynx → Oesophagus → Gizzard
- (3) Mouth → Gizzard → Crop → Pharynx → Oesophagus
- (4) Mouth → Pharynx → Oesophagus → Crop → Gizzard

Answer (4)

Sol. Correct sequence of foregut in cockroach is

Mouth → pharynx → Oesophagus → Crop → Gizzard.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 35 secs

Topic: Chemical Coordination and Integration

Concept: Endocrine Glands and their Hormones

Sub-concept: Hormones and their Chemical Composition

Concept field: Hormones and their Chemical Composition

52. Match the following columns and select the correct option.

Column-I

- (a) Pituitary hormone
- (b) Epinephrine
- (c) Endorphins
- (d) Cortisol
- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Column-II

- (i) Steroid
- (ii) Neuropeptides
- (iii) Peptides, proteins
- (iv) Biogenic amines
- (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Answer (2)

Sol. • Hormones of pituitary gland are protein hormones (peptides)

- Epinephrine is a Biogenic amine.
- Endorphins are neuropeptides.
- Cortisols are steroid in nature.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Animal Kingdom

Concept: Phylum Annelida

Sub-concept: Characteristic Features of Annelids

Concept field: Characteristic Features of Annelids

53. Which of the following options does correctly represent the characteristic features of phylum Annelida?

- (1) Triploblastic, unsegmented body and bilaterally symmetrical.
- (2) Triploblastic, segmented body and bilaterally symmetrical.
- (3) Triploblastic, flattened body and acoelomate condition.
- (4) Diploblastic, mostly marine and radially symmetrical.

Answer (2)

Sol. Phylum Annelida has the following characteristic features:

- Triploblastic
- Segmented body
- Bilateral symmetry

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Microbes in Human Welfare

Concept: Microbes as Biocontrol Agents

Sub-concept: Biological Control of Pests and Diseases

Concept field: Biological Control of Pests and Diseases

54. Match the following columns and select the correct option.

Column - I	Column - II
(a) Dragonflies	(i) Biocontrol agents of several plant pathogens
(b) <i>Bacillus thuringiensis</i>	(ii) Get rid of Aphids and mosquitoes
(c) Glomus	(iii) Narrow spectrum insecticidal applications
(d) Baculoviruses	(iv) Biocontrol agents of lepidopteran plant pests
	(v) Absorb phosphorus from soil

- (1) (a)-(iii), (b)-(v), (c)-(iv), (d)-(i) (2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
 (3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(v) (4) (a)-(ii), (b)-(iv), (c)-(v), (d)-(iii)

Answer (4)

Sol.

- Dragonflies are used to get rid of mosquitoes and aphids.
- *Bacillus thuringiensis* is used to control lepidopteran plant pests.
- Glomus (largest genus of arbuscular mycorrhizal fungi) absorbs phosphorus from soil.
- Baculovirus is a member of a family of DNA viruses infecting only invertebrate animals.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Digestion and Absorption

Concept: Alimentary Canal

Sub-concept: Oxyntic Cells

Concept field: Functions of Oxyntic Cells

55. Intrinsic factor that helps in the absorption of vitamin B₁₂ is secreted by

- (1) Goblet cells (2) Hepatic cells (3) Oxyntic cells (4) Chief cells

Answer (3)

Sol. Intrinsic factor that helps in absorption of vitamin B₁₂ is secreted by parietal cells (oxyntic cells).

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Chemical Coordination and Integration

Concept: Endocrine Glands

Sub-concept: Pituitary Gland and its Hormones

Concept field: Neurohypophysis and its Hormones

- 56.** Hormones stored and released from neurohypophysis are
- (1) Thyroid stimulating hormone and Oxytocin
 - (2) Oxytocin and Vasopressin
 - (3) Follicle stimulating hormone and Luteinizing hormone
 - (4) Prolactin and Vasopressin

Answer (2)

Sol. Hormones secreted by posterior pituitary (neurohypophysis) are:
Oxytocin and Vasopressin (ADH)

Class 12th

Question Type: NEET

Difficulty of question: Difficult.

Expected time to solve: 30 secs

Topic: Human Health and Diseases

Concept: Types of Diseases

Sub-concept: Causative Pathogen of Diseases

Concept field: Causative Pathogen of Diseases

- 57.** Match the following columns and select the correct option.

Column - I

- (i) Typhoid
- (ii) Malaria
- (iii) Pneumonia
- (iv) Filariasis

Column - II

- (a) *Haemophilus influenzae*
- (b) *Wuchereria bancrofti*
- (c) *Plasmodium vivax*
- (d) *Salmonella typhi*

- (1) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)
- (2) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a)
- (3) (i)-(a), (ii)-(c), (iii)-(b), (iv)-(d)
- (4) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)

Answer (1)

- Sol.**
- Causative pathogen of Typhoid is *Salmonella typhi*.
 - Malaria is caused by *Plasmodium vivax*.
 - Pneumonia is caused by *Haemophilus influenzae*.
 - Filariasis is caused by *Wuchereria bancrofti*.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 25 secs

Topic: Human Reproduction

Concept: Embryonic Development

Sub-concept: Features of Embryonic Development

Concept field: Features of Embryonic Development

58. In human beings, at the end of 12 weeks (first trimester) of pregnancy, the following is observed.

- (1) Eyelids and eyelashes are formed
- (2) Most of the major organ systems are formed
- (3) The head is covered with fine hair
- (4) Movement of the foetus

Answer (2)

Sol. In humans, by the end of first trimester (12 weeks) of pregnancy, most of the major organ systems are formed.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Neural Control and Coordination

Concept: Sensory Reception and Processing

Sub-concept: Eye

Concept field: Parts of an Eye

59. Match the following columns and select the correct option.

Column - I

Column - II

- | | |
|--|--|
| (a) Rods and | (i) Absence of Cones photoreceptor cells |
| (b) Blind Spot | (ii) Cones are densely packed |
| (c) Fovea | (iii) Photoreceptor cells |
| (d) Iris | (iv) Visible coloured portion of the eye |
| (1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv) | (2) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv) |
| (3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) | (4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i) |

Answer (1)

Sol.

- Rods and cones are the photoreceptor cells.
- Blind spot is the point where cone cells are absent.
- Fovea centralis is the point which has densely packed cones.
- Iris is the visible coloured portion of the eye.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Cell: Unit of life

Concept: Prokaryotic Cells

Sub-concept: Size of Prokaryotic Cells

Concept field: Size of Prokaryotic Cells

60. The size of Pleuropneumonia - like Organism (PPLo) is

- | | | | |
|-----------------------|----------------------|------------------------|----------------------|
| (1) $0.02\mu\text{m}$ | (2) $1-2\mu\text{m}$ | (3) $10-20\mu\text{m}$ | (4) $0.1\mu\text{m}$ |
|-----------------------|----------------------|------------------------|----------------------|

Answer (4)

Sol. $0.1\mu\text{m}$
PPLo are the smallest known cells.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Digestion and Absorption

Concept: Digestion of Protein

Sub-concept: Proteolytic Enzymes

Concept field: Proteolytic Enzymes

61. The proteolytic enzyme rennin is found in

- (1) Intestinal juice (2) Bile juice (3) Gastric juice (4) Pancreatic juice

Answer (3)

Sol. Rennin milk (protein digesting enzyme) is secreted by peptic cells found gastric mucosa. It is also called chymosin that curdles milk by converting soluble caseinogen into insoluble casein.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Animal Kingdom

Concept: Phyla of Animal Kingdom

Sub-concept: Characteristic Features of Different Phyla

Concept field: Characteristic Features of Different Phyla

62. Match the following group of organisms with their respective distinctive characteristics and select the correct option.

Organisms

Characteristics

- | | |
|--|---|
| (a) Platyhelminthes | (i) Cylindrical body with no segmentation |
| (b) Echinoderms | (ii) Warm blooded animals with direct development |
| (c) Hemichordates | (iii) Bilateral symmetry with incomplete digestive system |
| (d) Aves | (iv) Radial symmetry with indirect development |
| (1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) | (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) |
| (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii) | (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) |

Answer (1)

Sol. a-(iii): Platyhelminthes have only one opening in alimentary canal for ingestion and egestion, hence is incomplete.

b-(iv): Echinoderms (e.g. star fish) show radial symmetry and the larval stages are found in their life-cycle, hence, show indirect development.

c-(i): Hemichordates, known as half chordates. These are worm like tubular marine animals. Segmentation (also called metamerism) means linear series of repeating parts, this is not found in hemichordates. Their body is divided into proboscis, collar and trunk.

d-(ii) Aves (Birds) have stable body temperature and chicks grow into adults, no larval stages in life cycle. Hence, direct development.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 20 secs

Topic: Microbes in Human Welfare

Concept: Microbes in Medicine

Sub-concept: Bioactive Molecules

Concept field: Fungus *Trichoderma polysporum*

63. Cyclosporin A, used as immuno suppression agent, is produced from

- | | |
|--------------------------------|-------------------------------------|
| (1) <i>Monascus purpureus</i> | (2) <i>Saccharomyces cerevisiae</i> |
| (3) <i>Penicillium notatum</i> | (4) <i>Trichoderma polysporum</i> |

Answer (4)

Sol. Immunosuppressive agent cyclosporin A (obtained from fungus) destroys T-cell immune response in order to prevent organ rejection during organ transplantation.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 40 secs

Topic: Biotechnology – Principles and Processes

Concept: Separation and Analysis of DNA

Sub-concept: Biotechniques

Concept field: Biotechniques for DNA Processing

64. Select the correct statement from the following.

- (1) Gel electrophoresis is used for amplification of a DNA segment.
- (2) The polymerase enzyme joins the gene of interest and the vector DNA.
- (3) Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzymes of optimum conditions.
- (4) PCR is used for isolation and separation of gene of interest.

Answer (3)

Sol. Restriction enzymes are used to splice desired DNA segments, which are further inserted at specific locations of vector DNA.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Excretory products and their Elimination

Concept: Nephron

Sub-concept: Reabsorption

Concept field: Functions of Tubules

65. The increase in osmolarity from outer to inner medullary interstitium is maintained due to

- (i) Close proximity between Henle's loop and vasa recta
 - (ii) Counter current mechanism
 - (iii) Selective secretion of HCO_3^- and hydrogen ions in PCT
 - (iv) Higher blood pressure in glomerular capillaries
- (1) Only(ii) (2) (iii) and (iv) (3) (i), (ii) and (iii) (4) (i) and (ii)

Answer (4)

Sol. Increased osmolarity in medullary interstitium is essential for the formation of concentrated urine which there by facilitates effective water reabsorption. To achieve it, the Henle's loop and vasa recta are placed closely and the counter current developed thus affects the water reabsorption. The high pressure in glomerular capillaries helps in ultrafiltration helpful in increasing osmolarity.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 20 secs

Topic: Human Health and Disease

Concept: Immunity

Sub-concept: Types of Immunity

Concept field: Passive Immunity

66. The yellowish fluid "colostrum" secreted by mammary glands of mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. This type of immunity is called as

- | | |
|-----------------------|---------------------|
| (1) Passive immunity | (2) Active immunity |
| (3) Acquired immunity | (4) Autoimmunity |

Answer (1)

Sol. Passive immunity means when active antibodies from other's body are taken in. Colostrum has IgA that protects infants from ingested or inhaled pathogens.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Structural Organisation in Animals

Concept: Anatomy of Cockroach

Sub-concept: Digestion in Cockroach

Concept field: Digestion in Cockroach

67. Match the following columns with reference to cockroach and select the correct option.

Column - I

- (a) Grinding of the food particles
- (b) Secrete gastric juice
- (c) 10 pairs
- (d) Anal cerci

Column - II

- (i) Hepatic caecal
 - (ii) 10th segment
 - (iii) Proventriculus
 - (iv) Spiracles
 - (v) Alary muscles
- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(2) (a)-(iv), (b)-(iii), (c)-(v), (d)-(ii)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

Answer (1)

Sol. • **In Cockroach:**

- Grinding of the food particles takes place in proventriculus.
- Hepatic caeca secrete Gastric juice
- 10 pairs of spiracles are present.
- Anal cerci lies in the 10th segment.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Biotechnology – Principles and Process

Concept: Application of Biotechnology

Sub-concept: Pest Resistant Plants

Concept field: Pest Resistant Plants

68. RNA interference is used for which of the following purposes in the field of biotechnology?

- (1) to develop a plant tolerant to abiotic stresses
- (2) to develop a pest resistant plant against infestation by nematode
- (3) to enhance the mineral usage by the plant
- (4) to reduce post harvest losses

Answer (2)

Sol. RNAi process is involved in silencing of specific mRNA by formation of ds RNA. This prevents translation of the mRNA of the pathogen. Hence the pathogen does not survive in the host transgenic plants.

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 45 secs

Topic: Biotechnology – Principles and Processes

Concept: PCR

Sub-concept: PCR Rate

Concept field: Calculation of PCR Rate

- 69.** E.coli has only 4.6×10^6 base pairs and completes the process of replication within 38 minutes; then the average rate of polymerisation is approximately
- | | |
|----------------------------|----------------------------|
| (1) 2000 base pairs/second | (2) 3000 base pairs/second |
| (3) 4000 base pairs/second | (4) 1000 base pairs/second |

Answer (1)

Sol. 1 min = 60 sec.
38 min = 60×38 secs.,
= 2280 secs.
In 38 min $\rightarrow 4.6 \times 10^6$ bp
2280 sec $\rightarrow 4.6 \times 10^6$ bp
1 sec $\rightarrow \frac{4.6 \times 10^6}{2280}$ bp
 $\rightarrow 2017.54$
 ~ 2000 bp

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Reproductive Health

Concept: Contraceptive Methods

Sub-concept: Role of Steroids in Contraception

Concept field: Pills, Injections and Implants

- 70.** Progesterone alone or in combination with estrogens can be used as a contraceptive in the form of
- | | |
|------------------------------------|---------------------|
| (1) Implants only | (2) Injections only |
| (3) Pills, injections and implants | (4) Pills only |

Answer (3)

Sol. Progesterone used as contraceptive is in the form of progestin. It prevents ovulations and changes the mucus secretion in cervix, thus suppresses the phases of menstrual cycle.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Environmental Issues

Concept: Air Pollution and its Control

Sub-concept: Air Pollutants

Concept field: Size of Air Pollutants

- 71.** According to Central Pollution Control Board [CPCB] what size (in diameter) of particulate is responsible for causing greater harm to human health?
(1) 3.5 micrometres (2) 2.5 micrometres (3) 4.0 micrometres (4) 3.0 micrometres

Answer (2)

Sol. Particulate size 2.5 mm settles deep in lungs and cannot be coughed out. Thus, causes serious lung problems.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 35 secs

Topic: Breathing and Exchange of Gases

Concept: Respiratory Volumes and Capacities

Sub-concept: Respiratory Volumes

Concept field: Respiratory Volumes

- 72.** The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of a forced inspiration. This includes
(1) RV, IC (Inspiratory Capacity), EC (Expiratory Capacity) and ERV
(2) RV, ERV, IC and EC
(3) RV, ERV, VC (Vital Capacity) and FRC (Functional Residual Capacity)
(4) RV (Residual Volume), ERV (Expiratory Reserve Volume), TV (Tidal Volume) and IRV (Inspiratory Reserve Volume)

Answer (4)

Sol. TLC is 4-6 litres of air that includes:

RV = 1.2L	IRV = 3.3 L
ERV = 1.1 L	TV = 0.5 L

Class 12th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 35 secs

Topic: Human Reproduction

Concept: Gametogenesis

Sub-concept: Chromosome Number

Concept field: Chromosome Number

- 73.** Select the correct option of haploid cells from the following groups.
(1) Primary oocyte, Secondary oocyte, Spermatid
(2) Secondary spermatocyte, First polar body, Ovum
(3) Spermatogonia, Primary spermatocyte, Spermatid
(4) Primary spermatocyte, Secondary spermatocyte, Second polar body

Answer (2)

Sol. Secondary spermatocyte, first polar body and ovum, all are produced after meiotic divisions from diploid primary spermatocyte and diploid primary oocyte hence, are haploid.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Cell Cycle and Cell Division

Concept: Phases of Cell Cycle

Sub-concept: Meiosis

Concept field: Stages of Meiosis

74. During Meiosis I, in which stage synapsis takes place?

- (1) Pachytene (2) Zygotene (3) Diplotene (4) Leptotene

Answer (2)

Sol. Synapsis also called syndesis is pairing of homologous chromosomes that occurs during Zygotene (sub-phase) of meiosis I.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 35 secs

Topic: Cell – Unit of Life

Concept: Cell Organelles

Sub-concept: Functions of Cell Organelles

Concept field: Functions of Cell Organelles

75. Match the following columns and select the correct option.

Column - I

- (a) Smooth endoplasmic reticulum (i)
(b) Rough endoplasmic reticulum
(c) Golgi complex
(d) Centriole

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Column - II

- Protein synthesis
(ii) Lipid synthesis
(iii) Glycosylation
(iv) Spindle formation

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Answer (1)

Sol.

- Smooth endoplasmic reticulum synthesizes lipid.
- Rough endoplasmic reticulum synthesizes protein.
- Glycosylation takes place in Golgi complex.
- Centriole is related to spindle formation.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 35 secs

Topic: Excretory Products and their Elimination

Concept: Regulation of Kidney Function

Sub-concept: Role of ANF, Hormones in Blood Flow

Concept field: RAAS Mechanism

76. Select the correct statement.

(1) Atrial Natriuretic Factor increases the blood pressure.

(2) Angiotensin II is a powerful vasodilator.

(3) Counter current pattern of blood flow is not observed in vasa recta.

(4) Reduction in Glomerular Filtration Rate activates JG cells to release renin.

Answer (4)

Sol. Renin increases blood pressure in glomerulus by causing vasoconstriction in afferent arterioles of Glomerulus. Thus, required blood pressure in Glomerular apparatus is obtained.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Body Fluids and Circulation

Concept: Cardiac Cycle

Sub-concept: Cardiac Output

Concept field: Cardiac Output

77. Which of the following is associated with decrease in cardiac output?

- | | |
|------------------------|------------------------------------|
| (1) Sympathetic nerves | (2) Parasympathetic neural signals |
| (3) Pneumotaxic centre | (4) Adrenal medullary hormones |

Answer (2)

Sol. PNS (Parasympathetic neural signals) releases a hormone acetylcholine during excitement, stress etc., causing decrease in heart rate.

Class 12th

Question Type: NEET

Difficulty of question: Easy

Expected time to solve: 30 secs

Topic: Strategies for Enhancement in Food Production

Concept: Animal Breeding

Sub-concept: Inbreeding Depression

Concept field: Close Breeding

78. Inbreeding depression is

- (1) Reduced motility and immunity due to close inbreeding
- (2) Decreased productivity due to mating of superior male and inferior female
- (3) Decrease in body mass of progeny due to continued close inbreeding
- (4) Reduced fertility and productivity due to continued close inbreeding

Answer (4)

Sol. Inbreeding depression increases homozygosity that over the time causes expression of lethal genes. This reduces fertility and thus reduces production.

Class 11th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Structural Organisation in Animals

Concept: Connective Tissues

Sub-concept: Loose Connective Tissue

Concept field: Examples

79. Select the incorrectly matched pair from following.

- (1) Chondrocytes - Smooth muscle cells
- (2) ~~Neurons - Nerve cells~~
- (3) Fibroblast - Areolar tissue
- (4) Osteocytes - Bone cells

Answer (1)

Sol. Chondrocytes - cartilage cells.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 30 secs

Topic: Biotechnology and its Application

Concept: Ethical Issues

Sub-concept: Bio-patenting Biopiracy

Concept field: Bio-patenting Biopiracy

80. The laws and rules to prevent unauthorised exploitation of bio-resources are termed as

- (1) Biopatenting (2) Bioethics (3) Bioengineering (4) Biopiracy

Answer (1)

Sol. Bio-patenting means the patent holder exclude others for using the bio resources to produce any product of use. Patent holder's nod is required if patented bio-resource is to be used by a second party.

Class 12th

Question Type: NEET

Difficulty of question: Moderate

Expected time to solve: 35 secs

Topic: Human Reproduction

Concept: Male and Female Reproductive Systems

Sub-concept: Hormones Secreted by Reproductive Systems

Concept field: Hormones Secreted by Reproductive Systems

81. Match the following columns and select the correct option.

Column - I

Column - II

(a) Ovary

(i) Human chorionic Gonadotropin

(b) Placenta

(ii) Estrogen & Progesterone

(c) Corpus luteum

(iii) Androgens

(d) Leydig cells

(iv) Progesterone only

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Answer (4)

Sol.

- Ovary secretes estrogen and progesterone.
- Placenta secretes Human chorionic gonadotropin.
- Corpus luteum secretes only progesterone.
- Leydig cells secretes androgens.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 40 secs

Topic: Animal Kingdom

Concept: Phyla

Sub-concept: Animals in Different Phyla

Concept field: Common Names of Different Animals

82. Match the following columns and select the correct option.

Column - I

Column - II

(a) *Aptenodytes*

(i) Flying fox

(b) *Pteropus*

(ii) Angel fish

(c) *Pterophyllum*

(iii) Lamprey

(d) *Petromyzon*

(iv) Penguin

(1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Answer (3)

- Sol.**
- *Aptenodytes* is genus for Penguin.
 - *Pteropus* (bat) is also called as flying fox.
 - *Pterophyllum* is also called as Angel fish.
 - *Petromyzon marinus* is commonly known as lamprey.

Class 12th**Question Type:** NEET**Difficulty of question:** Difficult**Expected time to solve:** 35 secs**Topic:** Evolution**Concept:** Human Evolution**Sub-concept:** Cranial Capacity**Concept field:** Cranial Capacity

- 83.** A Hominid fossil discovered in Java in 1891 (now extinct), having cranial capacity of about 900 cc was
- | | |
|-------------------------|-----------------------------|
| (1) <i>Homo erectus</i> | (2) Neanderthal man |
| (3) <i>Homo sapiens</i> | (4) <i>Australopithecus</i> |

Answer (1)

- Sol.** *Homo erectus* fossil specimens are the first related members of genus *Homo*.

Class 11th**Question Type:** NEET**Difficulty of question:** Difficult**Expected time to solve:** 35 secs**Topic:** Cell Cycle and Cell Division**Concept:** Phases of Cell Cycle**Sub-concept:** Characteristics of Phases of Cell Cycle**Concept field:** Characteristics of Phases of Cell Cycle

- 84.** Match the following events that occur in their respective phases of cell cycle and select the correct option.
- | | |
|--------------------------|---|
| (a) G_1 phase | (i) Cell growth and organelle duplication |
| (b) S phase | (ii) DNA replication and chromosome duplication |
| (c) G_2 phase | (iii) Cytoplasmic growth |
| (d) Metaphase in M-phase | (iv) Alignment of chromosomes |
- | | |
|--|--|
| (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) | (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) |
| (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii) | (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) |

Answer (4)

- Sol.**
- Cell growth and organelle duplication takes place in G_1 phase.
 - DNA replication and chromosome duplication takes place in S phase.
 - Cytoplasmic growth takes place in G_2 phase.
 - Alignment of chromosomes takes place in Metaphase.

Class 11th**Question Type:** NEET**Difficulty of question:** Difficult**Expected time to solve:** 30 secs**Topic:** Breathing and Exchange of Gases**Concept:** Process and Site of Respiration**Sub-concept:** Regulation of Respiration**Concept field:** Regulation of Respiration

85. Match the following columns and select the correct option.

Column - I

- (a) Pneumotaxic Centre
- (b) O₂ Dissociation curve
- (c) Carbonic Anhydrase (iii)
- (d) Primary site of exchange of gases
- (1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
- (3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

Column - II

- (i) Alveoli
- (ii) Pons region of brain
- Haemoglobin
- (iv) R.B.C.
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

Answer (2)

Sol.

- Pons region of brain is known as Pneumotaxic center.
- Haemoglobin is related to O₂ dissociation.
- Carbonic anhydrase is responsible for reversible reaction for CO₂ and water.
- Primary site for exchange of gases is alveoli.

Class 12th

Question Type: NEET

Difficulty of question: Difficult.

Expected time to solve: 35 secs

Topic: Molecular Basis of Inheritance

Concept: Human Genetics

Sub-concept: Fingerprinting Process

Concept field: Fingerprinting Process

86. Which is the basis of genetic mapping of human genome as well as DNA fingerprinting?

- (1) Polymorphism in DNA sequence
- (2) Single nucleotide polymorphism
- (3) Polymorphism in hnRNA sequence
- (4) Polymorphism in RNA sequence

Answer (1)

Sol. DNA polymerisation means repeated DNA segments that are individual specific.

Class 11th

Question Type: NEET

Difficulty of question: Difficult

Expected time to solve: 30 secs

Topic: Body Fluids and Circulation

Concept: Blood Groups

Sub-concept: Rh Factor

Concept field: Disease Based on Rh Factor

87. Which of the following conditions cause erythroblastosis foetalis?

- (1) Mother Rh^{+ve} and foetus Rh^{-ve}
- (2) Mother Rh^{-ve} and foetus Rh^{+ve}
- (3) Both mother and foetus Rh^{-ve}
- (4) Both mother and foetus Rh^{+ve}

Answer (2)

Sol. When foetus is Rh positive and mother Rh negative, then mother's body start producing anti Rh bodies that reach the foetus and kill foetal RBCs. Thus it is lethal for the foetus.

Class 11th**Question Type:** NEET**Difficulty of question:** Moderate**Expected time to solve:** 30 secs**Topic:** Animal Kingdom**Concept:** Classification of Chordata**Sub-concept:** Characteristics of Chordates**Concept field:** Characteristics of Chordates

88. All vertebrates are chordates but all chordates are not vertebrates, why?

- (1) Notochord is replaced by vertebral column in adult of some chordates.
- (2) Ventral hollow nerve cord remains throughout life in some chordates.
- (3) All chordates possess vertebral column.
- (4) All chordates possess notochord throughout their life.

Answer (1)

Sol. Protochordates are the member of Chordata and in these notochord is not replaced by vertebral column. While in vertebrates, notochord is present during embryonic development which latter on replaced by vertebral column (back bone).

Class 11th**Question Type:** NEET**Difficulty of question:** Difficult**Expected time to solve:** 35 secs**Topic:** Locomotion and Movement**Concept:** Skeletal System**Sub-concept:** Disorders of Muscular and Skeletal System**Concept field:** Disorders of Muscular and Skeletal System

89. Match the following columns and select the correct option.

Column - I

- (a) Gout
- (b) Osteoporosis
- (c) Tetany
- (d) Muscular dystrophy

Column - II

- (i) Decreased levels of estrogen
- (ii) Low Ca^{++} ions in the blood
- (iii) Accumulation of uric acid crystals
- (iv) Auto immune genetic disorder

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(v)

(3) (a)-(iv), (b)-(v), (c)-(i), (d)-(ii)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Answer (2)

Sol.

- Accumulation of uric acid crystal in the body is symptom of Gout.
- Decreased levels of estrogen in menopausal women cause osteoporosis.
- Low Ca^{++} level in blood causes Tetany.
- Muscular dystrophy is an auto immune genetic disorder.

Class 12th**Question Type:** NEET**Difficulty of question:** Easy**Expected time to solve:** 15 secs**Topic:** Molecular Basis of Inheritance**Concept:** The DNA**Sub-concept:** Structure of Polynucleotide Chain**Concept field:** Nitrogenous Base

90. In the polynucleotide chain of DNA, a nitrogenous base is linked to the $-\text{OH}$ of

-
- (1) 3'C pentose sugar
 - (2) 5'C pentose sugar
 - (3) 1'C pentose sugar
 - (4) 2'C pentose sugar

Answer (3)

Sol. The nitrogenous base is linked to 1'C of pentose sugar.

